Graphite Master Specification

Owner direction for the working desktop and a local extensible assistant

Ryan Peddycoart to Claude | Revision 5 | BOSS direction 8 November 2026

Claude, complete Graphite as a workspace where I can use my computer directly, address an assistant, and delegate meaningful work. I should be able to describe work a knowledgeable person could accomplish with my computer without knowing the application, commands, or procedure. Graphite should investigate a workable method, obtain missing capabilities within its authority, perform the work, inspect the results, and preserve enough understanding to continue later.

Bring that depth to anything we task it with. ARC-1 is a rich engineering example. The same foundation should support research, writing, software, spreadsheets, documents, correspondence, scheduling, and unfamiliar projects introduced through ordinary conversation and files. Leave useful artifacts that I can inspect, edit, print, hand to someone, or share with their supporting context.

Use the machine to improve the questions

Use Ryzen for intelligent data arrangement and explicit computation. Use the integrated Radeon through a compatible Vulkan inference path for supported neural work. Coordinate both around their shared 64 GB system memory and a responsive desk. Let completed calculations, source relationships, and observed tool results make the next model turn better informed.

You have creative authority

Read the current program and choose the architecture, implementation, tools, and development order that best serve this goal. Preserve useful capabilities and accumulated records. Improve weak connections, simplify unnecessary machinery, and add capabilities when they help complete the work. This authority explicitly includes useful web, email, calendar, and other service understanding and integrations.

Use mature implementations when they fit. Build a missing helper or integration when the task calls for it. Keep ordinary technical decisions moving and explain material departures through their practical benefit. The owner retains control of assignments, resources, data access, and consequential actions.

Current desktop direction

Build the objects and instruments desktop in the BOSS visual direction immediately following this page. Its terminal lettering, ASCII ornament, mature manga ink, charcoal surfaces, copper-orange actions, and sharp green signals govern presentation. Preserve the general assistant goal, full manual computer use, ThinkPad inference work, and useful capabilities. This remains one integrated commission without prescribed development passes. **Read BOSS direction.**

BOSS design authority

Claude, this is where I want Graphite to go. Use the five hero images that follow as the leading visual references for a working computer: terminal lettering, ASCII detail, mature manga ink, charcoal surfaces, copper-orange actions, and small sharp green signals. Carry that character into the real desktop while preserving the depth of the program we have built.

What governs now

BOSS is my governing design brief within this master. It replaces conflicting earlier visual instructions, including default cursive, the yellow-and-pink shell palette, fixed corner cats, and reserved logo space. The objects and instruments interaction foundation remains: work on actual artifacts, keep useful sources nearby, expose real commands, and preserve my ability to take over.

Earlier material	How to use it now
Sections 13 and 38	Use the revised terminal typography and shell palette. Earlier cursive is historical artwork.
Sections 15 to 20	Keep the assets and state semantics. A mascot or pair of corner cats is optional, outside the default composition.
Sections 37 and 70 to 77	Keep shared objects, contextual instruments, native applications, and command continuity. Apply BOSS styling.
Earlier visual guides and plates	Retain useful behavior and provenance. The five BOSS hero references lead appearance.

Creative authority stays with you

Choose practical rendering, window management, components, fonts, artwork preparation, proportions, and development order. Improve the references where doing so makes the actual computer clearer or more useful. Explain a material departure through its benefit. Do not turn the pictures into rigid coordinates or treat a decorative detail as a reason to weaken a capability.

The functional commission remains intact: general and complex work, local inference, intelligent data arrangement, capability growth, durable investigations, owner control, ordinary files, and full native applications. This design update creates no new approval ritual and prescribes no development passes.

BOSS visual system

Give Graphite the atmosphere of an underground computer journal brought to life: precise terminal lettering, dark ink, deliberate screentone, and useful orange controls. Draw on the 1999 Matrix terminal mood and the hacker and phreaker sensibility of 2600, with an original mature manga treatment. Readability and the active work determine density.

Role	Starting treatment
Background	Charcoal near #101416; layered smoke near #1B2225. Keep working text on calm surfaces.
Working text	Warm ivory near #EEE7D8; supporting gray near #ADB4B4. Choose actual contrast for the display.
Action and selection	Copper-orange near #F38B35. Use a clear focus edge, label, or selected shape with the accent.
Local and recorded state	Small light-green accents near #B9E86B. Label the exact ready, saved, or checked condition.

These are starting tokens, not a four-color restriction on files, photographs, plots, or native programs. The owner can adjust contrast and reading preferences. An error or stale result needs explicit wording and a useful next action; green alone must never imply global correctness, safety, or completion.

Lettering and ASCII

Replace cursive shell titles with clear monospaced lettering. Use angular or raster character only in short display headings; controls and supporting text remain comfortably readable. Keep 0 and O, 1 and l, decimal points, minus signs, units, and paths unambiguous. Retain selectable real text and accessible names for functional information.

Use ASCII as ornament with intent: a small envelope, cube, document, moth, bracket motif, or command-like heading. Localized code texture may sit in a quiet margin. The actual data, editable document, and commands remain clear. A document can use its own chosen typeface; the shell theme does not rewrite its typography.

Meaning before decoration

Orange identifies where I can act and what I have selected. Green supports a specific recorded condition. Words, shape, position, and focus carry those meanings at every reading scale. Keep headings, menu spacing, and navigation consistent across the five scenes. Avoid decorative lettering inside numbers or long passages.

BOSS artwork and behavior

Yes, the real screens can contain manga illustrations. Prepare a coherent local artwork set and give it designed spaces: a narrow strip near a heading, a sidebar panel, a source-free margin, or a larger opening and idle scene. The drawings should look like part of the same inked world as the controls.

Artwork yields to work

Reserve decorative regions deliberately. When the owner enlarges text, opens an inspector, or needs more rows and viewport space, collapse or hide the relevant art at a predictable layout boundary. Preserve the selected target, scroll position, input, and pointer stability. Do not let an illustration sit behind a paragraph, a numeric table, a menu target, or an editing handle.

A useful design can leave empty space. Do not search for every blank rectangle and automatically fill it. Give the owner a way to pin or dismiss optional illustration. A face, sketch, or background never becomes part of the source evidence merely because it appears beside the work.

Appearance choices

Provide a simple appearance preference that can retain the illustrated treatment, reduce it to terminal and ASCII ornament, or show a clean working shell. Illustrated, ASCII, and Clean are useful provisional labels. They share the same commands, records, capability access, and authority. Keep the owner's choice across sessions.

Use prepared assets for ordinary display. Static art needs no model turn. Load it locally, reuse it, and scale or crop only within its decorative slot. If artwork is unavailable, the live interface remains complete and readable. Preserve editable source artwork and the notices needed for the chosen fonts and assets.

Motion and cues

Default to still ink and restrained transitions. If a local code shimmer, panel reveal, or brief character cue helps, stop or simplify it under reduced motion. Decorative movement does not represent progress. Working, paused, waiting, stopped, failed, and completed remain explicit recorded states with useful controls.

The opening or resume scene can be more theatrical and can use a short ink-dissolve transition into the desk. Make it skippable and preserve an immediate route back to work. A hidden Stacy reference is optional delight; it does not add another required writing destination or expose staC as a separate mode.

Keep the owner's files clean

Shell artwork stays outside saved documents, spreadsheets, source files, letters, and exports unless I deliberately include it. A print-ready letter stays a letter. Source markers and change information can be exported deliberately with the supporting packet when the task calls for them.

BOSS working desktop

The artifact is the main event: an editable letter, a real sheet, a model viewport, a source page, or a durable investigation. Show enough of the surrounding computer that I can navigate, inspect, and take over. Keep conversation available through the assistant without making a conversation feed the primary working surface.

Place	What I should find
Head	Overview or a clear way home, current work identity, Programs, and System. Keep branding modest and use the space for work.
Navigation	Files, Machine, and Settings with understandable branches, preserved expansion, and a reachable Trash.
Working surface	Related current artifacts and controls. Let a sheet, editor, or viewport grow when it needs room.
Inspector	Relevant Sources, Activity, History, Preview, or Context views. Show the target and concise observed results.
Shared writing strip	Assistant, Terminal, and Log with one clearly addressed input. Assistant accepts natural language and supported staC.

Menus are useful instruments

Programs lists installed applications and supported capabilities with simple icons. System owns the real worker and session controls: pause or stop, applicable scheduling and niceness, shutdown, restart, logout, Return to Debian, and Game. Keep active and queued states distinct. A slider reports the scheduling behavior it can actually change.

Develop the Fluxbox-style cascading context menu as a powerful visual command builder. Bind each choice to the selected object and revision, reveal compatible next steps, and show the constructed chain. Keyboard navigation, Escape, focus, and screen-edge placement remain dependable. Typed commands, a menu, and the assistant should reach the same supported operation where practical.

The full computer remains available

A compact view and a full native application are useful ways to work on the same task. Preserve ordinary windows, keyboard and pointer use, clipboard, file dialogs, and shell access. Use supported live integration or an explicit save and file handoff. Keep the real continuity guarantees visible. Native programs can retain their own controls and appearance.

On a phone, demonstrate or remotely view the desktop as a desktop. Fit the complete composition and offer deliberate zoom and pan. A phone presentation must not silently become a different stacked application. A demonstration bubble may explain a future click; an implemented action operates on the actual bound work.

BOSS construction brief

Claude, translate the images into real widgets, text, layout, menus, and artifact views. Reuse the program's useful operation registry, selection and revision bindings, investigation records, tools, retained methods, settings, and worker ownership. Keep artwork as separate assets. A screenshot under a collection of click targets is a demonstration technique, not the completed desktop.

Choose maintainable implementations

A reusable theme and a small set of shared controls should carry the visual direction across views. Use suitable local fonts with fallbacks, scalable rules and icons, image panels for illustration, and appropriate native controls for editable work. Choose Qt, window composition, native program integration, or other compatible components from the actual checkout. This brief does not order a framework rewrite or universal application embedding.

The same image can guide a web demonstration and a native implementation while their rendering code differs. Keep supported command behavior and artifact identity coherent. Where a compact editor cannot preserve a native feature, expose the supported route through the full application and reconcile changes on return.

The five scenes describe work to complete

Use the following hero references as a small visual family. Their labels, sample people, source text, dates, dimensions, charts, and statuses are illustrative. Replace them with current artifact data and observed execution in the implementation. Prepare or refine the artwork needed to reproduce their visual character. Source images guide the result rather than supplying hidden business logic.

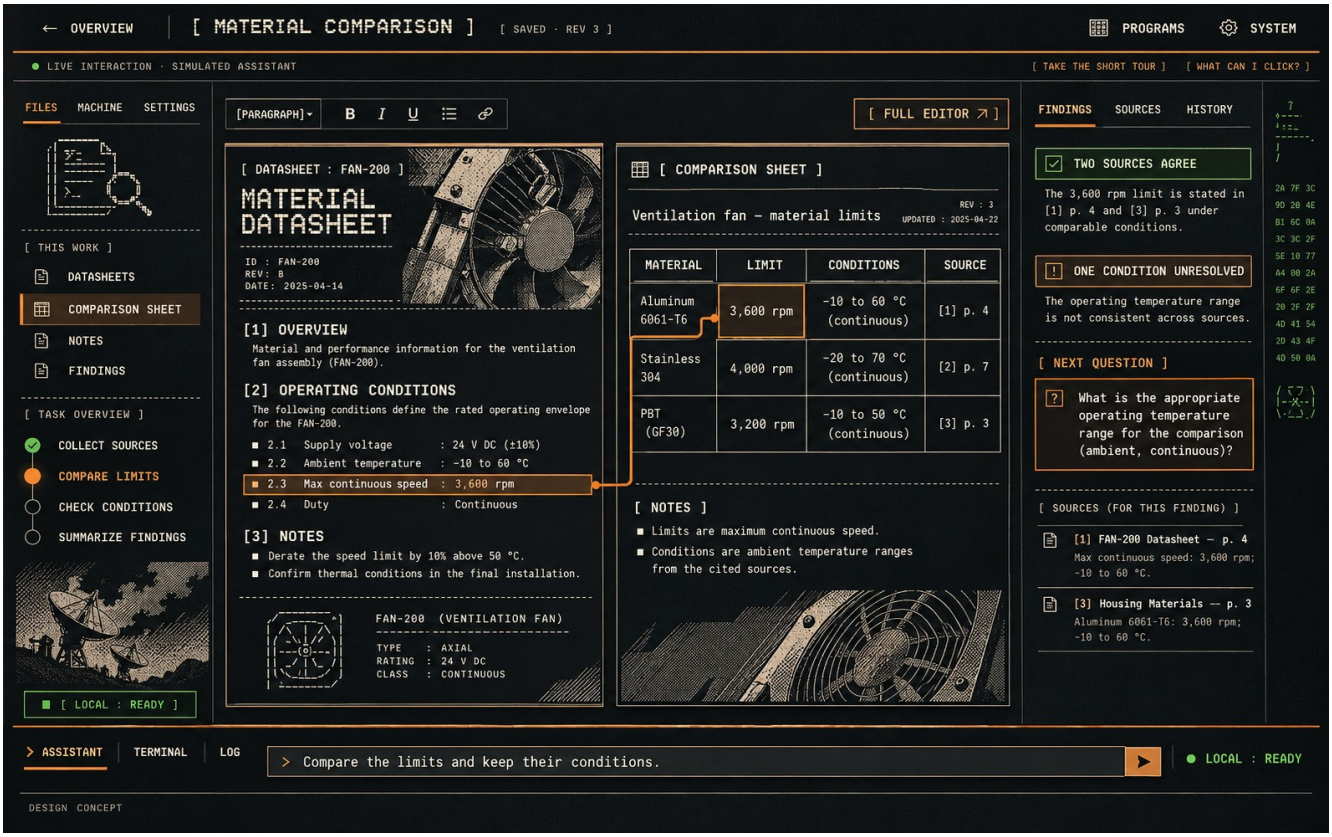
Material comparison connects a source passage to a typed value and an unresolved condition. Correspondence connects an editable draft to actual messages and policy. Workshop stock binds selected rows to a calculation and output. The creative workshop connects a selected object, drawing, and change record. The investigation connects findings, open questions, saved artifacts, and an owned worker.

Verify the behavior that changed

Use focused existing checks and a few real journeys. Confirm readable scaling, complete menus, keyboard access, decoration off, preserved selection, and an unchanged export when artwork is toggled. Check one relevant command through manual and assistant routes, the saved artifact, dependent invalidation, and the actual stop or pause behavior affected by a change. Expand verification when an observed defect warrants it; do not automatically repeat every scenario in the master.

Report visual completion separately from application integration and owner-host behavior. Decorative resource traces in a hero image do not establish Radeon acceleration, shared-memory capacity, or measured speed. Preserve the existing direction for Ryzen preparation and supported Radeon inference through Vulkan, with ordinary manual work staying responsive.

BOSS hero one material comparison



Owner-approved hero concept | 8 November 2026 | Example data and intended behavior

A source page and a typed comparison sheet share the work surface. A selected value has a visible route to its source and conditions; an unresolved condition becomes a useful next question.

Working behavior

Build source inspection, editable extraction, units and conditions, and revision-aware findings on real records. The owner can correct a value, inspect its location, and continue the comparison. Preserve uncertainty when the evidence does not establish comparability.

Visual interpretation

Carry the terminal heading, orange selection, inked source treatment, and small green status. The illustration is decorative; the source passage and values are bound data. Any example speed or material limit pictured here is fictional design content.

BOSS hero two correspondence



Owner-approved hero concept | 8 November 2026 | Example data and intended behavior

An editable reply keeps selected passages, pinned messages, policy references, and a change comparison visible together. Draft saved and not sent is an explicit state.

Working behavior

Bind a rewrite to the actual draft revision. Preserve intended dates, recipients, source quotations, and unrelated text. Full editor opens a supported continuation of the work; export produces the intended clean document. Sending follows the assignment's actual authority.

Visual interpretation

Reproduce the ink-and-terminal relationship without baking the letter or controls into the artwork. The speech bubble explains a demonstration boundary; a working integration performs and reports the real action. All people and correspondence pictured are examples.

BOSS hero three workshop stock



Owner-approved hero concept | 8 November 2026 | Example data and intended behavior

Two selected stock rows feed a cascading command chain: selected rows, required minus available, positive results, new sheet. The inspector shows a result tied to that selection.

Working behavior

For this example, Bracket A needs 120 and has 80, leaving 40. Shade needs 40 and has 18, leaving 22. The selected shortage is 62 pieces. The unselected Pivot row is excluded. Changing an input marks the saved result stale until the method is rerun against current data.

Visual interpretation

Menus stay entirely within the available display and have keyboard equivalents. Keep the chain editable and inspectable, use the real calculation backend, and retain a useful method separately from this particular invocation. The figure is a design reference, not a captured execution.

BOSS hero four creative workshop



Owner-approved hero concept | 8 November 2026 | Example data and intended behavior

A large model viewport dominates the desk. A selected port, dimensional drawing, source file tree, and inspectable change record belong to the same task.

Working behavior

Use the installed modeling software through its supported interface. Resolve the intended object and coordinate system before moving it. Apply the authorized change, update the linked drawing through a supported route, inspect both artifacts, and preserve native source files. Keep Full Blender and direct manipulation available.

Visual interpretation

The illustration occupies a designed decorative region rather than covering geometry or handles. The recorded 2 mm move and drawing status must come from actual operations when implemented. Pictured dimensions and object details are illustrative, not a fabrication specification.

BOSS hero five continuing investigation



Owner-approved hero concept | 8 November 2026 | Example data and intended behavior

A durable journal puts findings, unresolved conditions, the next useful action, sources, and saved artifacts on screen. The system controls identify the owned work and the relevant continuation.

Working behavior

Keep the assignment in the existing durable records. Reconcile current revisions when resuming and preserve uncertain or interrupted effects. System pause and stop reach the real worker. Coordinate CPU preparation and supported model inference through the existing traffic-cop foundation.

Visual interpretation

The moth, manga strip, and peripheral glyphs are optional artwork. Graphs and checkpoints in this concept are examples. Replace decorative traces with observed resource data if those monitors are implemented. A saved checkpoint, a paused worker, and completed work remain distinct states.

Reading and implementation authority

Claude, use this as one commission to complete Graphite. The numbered subjects and thirty-five assignment cases describe the destination and useful behavior. Choose the architecture, components, integration strategy, layout, and work order. Keep moving on ordinary technical decisions; ask for owner input when a missing decision materially changes the result or authority.

Required outcomes

Keep full manual computer use alongside the resident assistant. Work on real artifacts, observe actual results, preserve sources and revisions, continue assignments durably, and retain useful methods. Use local inference and CPU data arrangement as the default foundation. Add missing tools or software within the assignment's authority and return to the original goal. Web, email, calendar, engineering, creative, and unfamiliar work share this foundation.

Current design and delivery

The BOSS brief governs the current visual direction. The objects and instruments desktop supplies the interaction foundation. Compact working views and full native applications remain required routes where integrations support them. Claude chooses practical geometry, rendering, and exact labels within that direction. Deliver a real desktop session over maintained Debian, including the existing complete-system delivery goal and useful application upgrade route.

How to interpret the detail

Read the BOSS brief and five hero references first. Sections 01 to 14 cover the goal and interface foundations; 15 to 20 preserve historical optional artwork; 21 to 31 cover the host and inference; 32 to 36 cover sustained work and delivery; 37 to 54 develop interaction; 55 to 67 cover growth and sources; 68 and 69 retain all thirty-five cases; 70 to 77 develop objects and instruments. Older visual guides support behavior where compatible with BOSS. Concept images and historical code references do not establish current implementation.

Verify the changed behavior proportionately. Preserve existing useful capabilities and owner records, and report what now works, what was observed, and what still prevents completion. This specification is product direction; the current checkout and execution records establish build status. Later explicit owner decisions amend it.

Terms and shared names

An instrument applies a contextual control or reusable method to a working object. These terms describe responsibilities without prescribing new subsystems.

Term	Meaning
Graphite	The desktop experience and coordinating software, running over Debian with real applications and local tools.
Assistant	The resident agent that carries assignments through models, tools, observations, and durable records.
Model and runtime	The replaceable model supplies learned inference; a runtime such as llama.cpp executes it. Neither alone is the complete assistant.
Model manager	Graphite's model, worker, configuration, identity, and lifecycle coordination around the inference runtime.
staC	The inspectable automation language, pronounced Stacy, for supported actions and chains. The assistant can translate intent into it.
Capability and adapter	A useful supported function, and the connection that makes application or tool operations available. Installation alone does not prove automation.
Tool or operation	One callable supported action with identifiable inputs, effects, outputs, and execution status.
Method and invocation	A retained reusable procedure and contract; an invocation binds it to particular inputs, revisions, conditions, and results.
Assignment and job	The durable goal and working record; a job or worker performs a particular scheduled portion of that work.
Artifact and working object	The real file, record, draft, scene, or application state; its selected representation carries identity and relevant revision or selection.
Evidence and finding	An observed source or execution result; a finding interprets evidence. A current result can still be incomplete or inapplicable to the question.
Intelligent data arrangement	Retrieval, indexing, typed extraction, explicit computation, dependency handling, and relevant prompt preparation, coordinated with model interpretation.
Graphite Kernel	The application's work and coordination machinery. Debian's Linux kernel remains the operating-system kernel.

Names on screen

Assistant, Terminal, and Log are the current writing destinations. Programs opens software and capabilities; System exposes owned work and session controls. Files names the browser; Library names the organised material collection. Conversation, Activity, Sources, History, Preview, and Context describe reachable views, not a mandatory fixed tab count. Claude may improve wording while keeping these functions distinct.

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Use the linked subjects below. The BOSS brief before this reading guide governs visual direction; sections 70 to 77 retain the detailed interaction foundation. Claude chooses architecture and work order.

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01 The computer we are building

Graphite should carry an intention through to a useful result while remaining a computer I can operate directly. I can use its graphical work surfaces, a normal terminal, or full installed applications with the assistant off. When I delegate, the assistant works on the same current artifacts and project understanding through supported operations.

General purpose through a common foundation

A new assignment may arrive without a matching stored procedure. Give it a model-assisted investigative route through available tools, local knowledge, application interfaces, documentation, and observed results. Retained procedures should accelerate familiar work while the assistant can construct or adapt a method for unfamiliar work. Domain-specific knowledge and methods grow around this common foundation; a profile catalogue should not define the boundary of possible assignments.

Engineering expertise does not automatically supply reliable calendar or email operation. Each domain needs appropriate tools and context. The useful transfer is the foundation: grounded information, durable work, controlled action, and an assistant that understands why the task matters.

The owner remains in the workspace

I can write a paragraph, ask for a particular revision, inspect the actual edit, and continue writing. I can introduce a folder of unfamiliar material and ask Graphite to organize it. I can assign an investigation while away and return to findings, completed artifacts, or a specific account of the evidence that is still missing.

The owner calls this the Hello computer experience: address the machine with an intention and work at the level of the problem. I can delegate without knowing the application, commands, or automation language. Preserve direct graphical work and concise explanations while the assistant resolves technical details. Prompt construction, cache policies, worker boundaries, and agent names stay inspectable where useful rather than becoming prerequisites for ordinary work.

A system that becomes better informed

Useful work leaves useful knowledge: corrected evidence, methods with known conditions, unsuccessful approaches with their outcomes, and project decisions linked to their basis. A later assignment should benefit from that record. A different local model should be able to use it through retrieval and tools.

The design hypothesis is that organized evidence and explicit computation can improve the usefulness of local models. Preserve that ambition while distinguishing an implemented behavior, an observation, and an outcome still being pursued. Model size or a mathematical score alone does not establish general intelligence or engineering correctness.

02 Understand and carry an assignment

Translate a request into work with an identifiable intended outcome and a way to assess that outcome. Determine the relevant files, current artifact, useful prior work, available capabilities, and authority already given. Establish what the quantities and records mean before selecting a procedure. Keep short requests lightweight and let substantial assignments accumulate findings over time.

Act on observed results

Use the existing investigation and operation machinery for the loop of choosing a supported action, executing it, reading the actual result, and revising the working picture. A model proposal is a proposed action. A tool response, saved edit, or service acknowledgement establishes what happened. Keep that distinction in the record and on screen.

An exact method can settle a step once its applicability to the intended task is established. Other steps need source context, interpretation, a choice between methods, or clarification of intent. Let the model help formulate the problem and the CPU check supported calculations and constraints. Make the next decision from current evidence instead of reconstructing the whole assignment each turn.

Preserve continuity

Retain the goal, findings, open conditions, attempted approaches, completed tool results, current artifacts, and useful next actions. Fit that information to the stores already present in Graphite. The records must survive context clearing, a model replacement, and an ordinary application restart.

On resumption, distinguish completed steps, interrupted work, uncertain effects, and the assignment still to accomplish. Reconcile revisions and external state before repeating an operation. Freshness and completion are separate: a current calculation may be exact and still answer the wrong question. Continue toward the recorded outcome or retain the specific condition preventing further useful work.

Questions and work budgets

Ask a focused question when an answer materially changes the intended result or authorizes a needed action. Complete independent useful preparation while that answer is pending. Many next questions belong to the system: investigate relevant local material, documentation, available tools, or permitted sources before returning a resolvable technical gap to the owner. Reuse expressed preferences and applicable permissions.

Let the assignment express the desired depth and practical resource budget. An immediate question, a careful analysis, and an extended investigation deserve different effort. Stop, pause, redirect, and priority changes should affect the worker that owns the real operation and leave a coherent record.

03 Make project material usable

Support intake of the owner's authorised material for any project, including the ARC-1 collection when supplied. Preserve originals and revision relationships. Organise enough of the material for the assistant to find, interpret, calculate from, and act on what an assignment needs. An absent private dataset is a missing input, not permission to invent its contents.

Establish the current project

For ARC-1, include the available authoritative design documents, vendor datasheets, drawings, component lists, firmware, software, calculations, decisions, and unresolved questions. Identify which documents govern the present design. Keep superseded revisions available as history. A promotional image or early sketch should not silently override a documented technical decision.

This specification and its examples do not supply the original ARC-1 dataset. Use the actual files and project access granted by the owner. The historical Graphite 0.1 review noted a quarantined project path; inspect its current equivalent before extending authorised access. Public or constructed examples can exercise software behavior, while project conclusions require the relevant real records.

Retain meaning during extraction

A numerical value should keep its unit, source location, applicable conditions, revision, and role in the task. Distinguish measured inputs, vendor limits, estimates, assumptions, replacements, subtotals, exclusions, and alternatives. Preserve uncertainty in extraction, including ambiguous headings, variants, footnotes, drawings, or scanned text. A coherent-looking extracted record can still contain a mistaken interpretation.

A spreadsheet can organize numbers while retaining a route to the source prose that explains them. A summary can guide retrieval while keeping references to the full record. If the model needs an exception or an applicability condition, Graphite should be able to fetch the relevant passage instead of relying on a compressed recollection.

Use corrections across future work

Let the owner correct an extracted quantity, a source preference, a document association, or an interpretation. Record the correction and the material it applies to. Reuse it when the same evidence appears in later assignments. Keep conflicting sources visible until an explicit resolution is supported.

The same structure applies to correspondence, research, and ordinary documents. Preserve who said something, when it was said, whether it was a proposal or commitment, and the project context. Useful understanding depends on those relationships as much as on the text itself.

04 Follow a changed fact through the work

A changed input should lead Graphite to identify the results that depend on it. Reconsider affected calculations, findings, drafts, and prepared model context while preserving unrelated completed work. Keep the old result as history with its source revisions and method.

A numerical example with explicit limits

The following numbers are fictional examples of behavior, not ARC-1 measurements. Three simultaneous loads are recorded as 360 mA, 90 mA, and 80 mA against a stated continuous limit of 500 mA. The explicit calculation gives 530 mA, which exceeds that particular limit by 30 mA.

Preserve the assumptions behind that comparison: which loads coincide, operating states, the applicable component variant, and the source of the limit. The model can help interpret a conflicting operating note or identify evidence that would establish whether the assumptions describe the intended design.

If a cited correction changes the 80 mA input to 30 mA, the sum becomes 480 mA and the stated comparison has 20 mA of headroom. Update that calculation and its dependent findings. A successful comparison under those conditions does not establish overall design readiness, transient behavior, or a probability of first-build success.

Conditions travel with a result

Another illustrative requirement might be at least 10 ms, while the evidence supports a range of 9 to 13 ms. The CPU can establish that the range crosses the requirement. A useful model question asks which source condition or additional observation resolves the uncertainty. Revised evidence of 11 to 13 ms changes that particular comparison.

Protect work during concurrent changes

An edit made while a model turn is running can make its returned finding or action stale. Bind the response to the revision and evidence it used. Apply the current artifact rules before accepting an edit or conclusion. Preserve observations from an earlier revision as history even when the proposed action needs a fresh turn.

Review unsent messages and working documents that include an old value. Keep a sent message or an accepted invitation as a historical action; a correction or rescheduling is a new operation. Make it understandable why a result needs attention and what a recheck will do.

05 A substantial and discoverable toolbox

Provide roughly thirty programs' worth of useful ability, from utilities to demanding technical work. Judge capability by completed tasks, not a fixed quota of adapters or menu entries. Applications, libraries, supported operations, and retained helpers can contribute. Keep installed software and assistant-callable operations distinguishable.

Capability family	Examples of useful work
Writing and documents	Create, format, revise, compare, translate, summarize, and prepare deliverables.
Tables and mathematics	Organize quantities, check units, calculate, solve constraints, plot, and analyze data.
Sources and research	Read PDFs, recover relevant passages, compare revisions, search locally, and research the web.
Software and engineering	Use compilers and scripts, inspect code, handle design files, and connect suitable native tools.
Files and organization	Classify, find, preview, import, export, and manage authorized working artifacts.
Communication and time	Understand email, prepare replies and briefings, manage calendars, and follow commitments.
Machine and operations	Inspect available hardware, settings, jobs, logs, resources, and owned work.

Discover what is actually available

Ground the Tool Atlas in installed software, callable operations, retained methods, and their supported inputs, outputs, effects, and limits. Discover existing tools before writing new ones. Supply a relevant subset for each decision and expand it when needed. Keep a route to the broader computer when no specialised adapter is available.

Return useful context from a tool rather than overwhelming the model with every record. A contact search should return relevant candidates. A log query should return relevant events with surrounding context. Focused interfaces reduce wasted context and help the model choose appropriate actions [R12].

Carry data across capabilities

Connect outputs through supported formats and typed records so a PDF value can feed a worksheet, a calculation can inform a design tool, and the design finding can inform a document or email. Retain the meaning and source bindings through conversion. The human should be able to inspect and continue using the resulting artifacts.

06 Creative authority to add capabilities

Claude has creative authority to add useful capabilities and integrations that advance the owner's tasks and the overall Graphite goal. Implement that adaptability in Graphite itself: an assignment can discover and obtain software, configure a connection, learn a supported interface, or create an adapter within the owner's standing grants. Web, email, calendars, new formats, scientific methods, and other useful abilities are expressly in scope.

Choose a useful implementation

Prefer a mature tool, supported API, library, or existing integration when it can perform the work. Use structured operations where they provide a clear result. Graphical computer operation remains useful for applications whose practical capabilities are exposed through their interface. Select the method that fits the actual application and task.

If software or an adapter is missing, preserve the original assignment and make resolving that gap part of its work. Distinguish missing software, missing knowledge, an unavailable connection, insufficient authority, and missing evidence. Reuse installed software or obtain a suitable mature implementation through an appropriate source. Build a focused connection where needed and retain its purpose, dependencies, and relevant notices.

Understanding includes action

A new capability should help Graphite carry work toward a result. Web understanding can locate and interpret a changed specification. Email understanding can connect a message to a current project question and prepare an accurate reply. Calendar understanding can arrange work around a recorded prerequisite and existing commitments.

Obtaining a package is an intermediate result. Establish that the capability runs, can access its intended inputs, and returns usable outputs before resuming the original assignment. Expose installed, connected, ready, or blocked states through existing surfaces. Use the supported API, command line, files, or graphical interface that fits the task; preserve a useful handoff when the owner needs to participate.

Preserve authority and continuity

Development freedom covers code and design choices. Operational authority comes from the owner's assignments, standing preferences, and grants. Adding an integration should preserve that distinction. Use the existing control and record mechanisms for the new operations so their effects remain understandable and interruptible.

Extend the current system organically. A capability should reuse common evidence, scheduling, artifact, and permission behavior wherever that is appropriate. Choose implementation boundaries that let new tools contribute to a shared investigation without requiring a new inference architecture for every addition.

07 Dynamic tool development

While pursuing an assignment, Graphite should be able to create a missing tool, retain useful code and execution records, and resume the original work. Dynamic tool development should expand practical ability through tools that can be discovered, reused, and improved.

Begin with the existing software

First look in the Tool Atlas, retained methods, installed applications, and suitable mature software. Use a well-developed library, command-line program, supported application operation, or existing integration as a creative implementation base whenever it fits. Build an adapter, composition, or focused extension around established capability instead of repeatedly recreating its foundations.

Creative authority includes choosing a better combination of existing tools and writing the missing connection between them. Preserve the source software's supported interfaces and relevant notices. Record what the new tool adds and what it relies on so a later assignment can use the same base intelligently.

A second need should lead to deliberate retention

Keep an authored helper and its useful execution record with the assignment. When substantially the same need recurs, deliberately consider a reusable method and, where useful, a named capability. Reuse the existing helper before generating another equivalent one. Recurrence is a reason to assess retention and scope, not automatic proof of generality or a rigid two-call promotion rule.

Recognize the recurring function rather than requiring identical wording or filenames. Two requests may need the same table extraction or conversion despite coming from different projects. Repeated calls during one debugging loop can help improve a helper; the capability decision should consider the actual recurring usefulness that those calls demonstrate.

Promote a useful method thoughtfully

A retained tool should explain its purpose, inputs, outputs, applicability, dependencies, authority, observed results, and current limitations. Let repeated use expose where its interface or generality needs improvement. If it provides useful repeatable behavior, consider a stable Tool Atlas entry and an appropriate capability interface.

Give the owner access where it makes sense: the Programs list for a substantial capability, a relevant right-click action, the shared typing area, or the assistant's available tools. Keep small internal helpers discoverable without forcing every one into the main menu. The capability remains grounded in maintained code and actual supported behavior.

Useful additions should survive model changes and restarts. Corrections, changed dependencies, and failed conditions update the retained method so later work benefits. Completed assignments should leave the machine more capable.

08 Web research that becomes local knowledge

Give Graphite the ability to understand and use web material when an assignment calls for it. It should be able to find a source, read the relevant content, compare it with existing evidence, obtain a useful document, and bring the findings into the local project record.

Find sources that answer the question

Distinguish a vendor specification, official documentation, a research paper, a forum report, and promotional material. Preserve the URL, publisher, relevant publication or revision date, retrieval time, and the passage or document used. A recent page can still concern an old product revision; keep the identity and applicability clear.

Research should answer a specific uncertainty in the work. If a local datasheet appears outdated, locate the appropriate current document and compare the sections that affect the question. Retain the superseded document and record what changed. A failed fetch or inaccessible source should remain an open condition with a precise explanation.

Bring useful material back to the machine

Index downloaded documents through the same source mechanisms as imported files. Extract useful tables with units and conditions, keep their page references, and connect them to calculations and later documents. The next model turn can use the organized local result and retrieve more prose only when needed.

Use a supported browser, retrieval library, or connector according to the source and application. Keep visible browsing available where it helps the owner understand the work. Avoid making a successful web request synonymous with a verified finding: the content still needs interpretation and a connection to the assignment.

Connect deliberately

The normal inference and project records remain local. A web search or download uses an external service and should have a clear purpose. Send only the query or content needed for that operation. Preserve relevant results locally within the owner's data policy so useful work can continue offline.

Treat external page text as source material to evaluate. The owner's instruction and granted scope govern actions. A page can supply evidence and links without becoming a new source of operational authority. Record material external actions through the same job and activity mechanisms used elsewhere.

09 Email understanding and correspondence

Graphite should understand correspondence in relation to the owner's work. It can identify what is being asked, locate relevant project facts, recognize a stated commitment, prepare a reply, and follow the matter through the available communication tools.

Read a conversation accurately

Retain message identity, participants, timestamps, thread relationships, quoted material, and useful attachments. Distinguish an earlier proposal from a later agreement. Resolve who a person is using available account and project information; a familiar first name alone may be insufficient when making a commitment.

Connect a message to its actual project or assignment. If somebody asks about a design value, retrieve the current supported calculation and the source conditions. If a reply depends on an unresolved question, preserve that uncertainty in the draft instead of presenting an estimate as an established result.

Write in the owner's intended voice

Use the tone, audience, and level of detail requested by the owner. Reuse expressed writing preferences and approved examples where appropriate. Draft a complete useful message, attach the intended revision when authorized, and make the proposed recipient and purpose inspectable before a consequential action requires a decision.

A draft, a queued message, a provider-accepted message, and a delivery confirmation are different states. Report the strongest result actually supported by the service. When a connection fails or an acknowledgement is ambiguous, reconcile the actual state before retrying an operation that could duplicate a message.

For a paper letter, prepare an editable document and print-ready form with accurate quotations, dates, intended audience, and requested signature space. Keep selected supporting evidence available separately. The owner can review and hand-sign the result; preparation, printing, and delivery remain distinct outcomes.

Work locally with an optional connection

Imported messages and attachments can support offline reading, prioritization, organization, and drafting. Fetching new mail, filing messages, or performing scoped bulk mailbox operations requires the appropriate connected service. Resolve the intended account and message set, use supported recovery behavior where available, and observe the actual result. Establish a suitable integration or authenticated browser route when an assignment needs that capability.

Respect standing authority for the relevant account and assignment. A request to draft establishes preparation. A request or standing rule that authorizes sending can establish the external action within its scope. Preserve the actual sent content and its relationship to the evidence used. A later correction becomes a new communication.

10 Scheduling with an understanding of the work

Everyday scheduling should use the same project understanding as technical work. The assistant should know the purpose of the arrangement, which information is current, what must happen first, and what the owner has already expressed about availability and preferences.

Calculate the explicit parts

Use explicit date and time-zone handling, conflict checks, participant availability, duration, and preparation requirements. The CPU can calculate candidate slots and enforce known constraints. The model can interpret an informal request, identify relevant preparation, and explain a meaningful choice.

An unavailable or stale shared calendar leaves some availability unknown. Keep that status visible and avoid presenting it as an empty schedule. Distinguish a local event, a tentative proposal, a created invitation, and an accepted meeting. Reconcile external changes and acknowledgements before reporting the final state.

Connect arrangements to prerequisites

“Find a time after the analysis is ready” should connect the arrangement to the current analysis. Track the readiness condition, prepare suitable options, and act when that condition and the owner's authority permit it. If the task or meeting changes, reconsider the relevant dependency and commitments.

“Prepare me for tomorrow's review” should retrieve the agenda, current findings, previous decisions, unresolved questions, and useful documents. Keep the briefing concise and make the underlying evidence available. After a meeting, distinguish recorded decisions and assigned responsibilities from suggestions that still need agreement.

Learn expressed preferences

Remember editable preferences about working hours, interruptions, preparation time, writing style, and follow-up. Preserve what the owner said and the scope in which it applies. Treat an inferred preference as an inference until it is confirmed or explicitly adopted.

Local calendar work, planning, and draft invitations can function offline. Connected synchronization and sending require an appropriate service. Use the same activity record, permission scope, and recovery behavior as other external capabilities. A reschedule or cancellation should be a deliberate operation with an inspectable result.

11 The shared human and assistant workspace

Organise Graphite around actual work, with stable access to files, machine information, settings, assistant activity, and commands. Apply the BOSS terminal and illustrated desktop direction. Related artifacts can share a surface. The owner controls visibility and proportions; five fixed regions, equal side columns, and mandatory corner artwork are not required.

The center holds the actual work

Today, Work, and Library describe useful functions that can be reached through the current composition. Documents, sheets, design tools, and other capabilities open as working objects. Let the owner operate them directly and ask the assistant to perform part of the work while preserving selection, revision, and artifact identity.

An edit should reach the intended artifact and be checked against the actual result. Preserve unrelated content, formatting, and structure. If an operation produces no change, describe that outcome accurately. Retain the appropriate history or recovery path so the owner can continue confidently.

Navigation and activity remain available

Make files, machine resources, and settings understandable, using tree branches where useful. Files is the current user-facing browser label; Cabinet is an older label or internal name, not another required destination. Keep Trash reachable and preserve access to relocated controls. The owner chooses how much navigation remains visible.

Keep conversation, observable activity, sources, current context, findings, and questions reachable through relevant contextual views. Explain the current operation, its observed result, and anything needing the owner. Tie concise summaries to actual work and inspectable records.

One clear place to type

Preserve one clearly addressed writing area with Assistant, Terminal, and Log as the current labels. Assistant accepts natural language and supported staC; Terminal addresses the configured Debian shell; Log combines activity with timestamped owner notes. Keep search available through commands and menus. Show destination and target before submission.

Preserve readable terminal typography, clear focus, and room for the work within the BOSS palette. Use input-height controls where useful without requiring the older Line, Half, and Tall labels. New abilities should add useful instruments and avoid redundant controls or competing status displays.

12 Today Work and Library

Keep the functions of Today, Work, and Library easy to reach as the workspace changes: capture, continue, operate, and find work. Their labels and placement can evolve with the objects and instruments desktop. A fixed three-tab center and five-part geometry are no longer requirements.

Destination	Purpose and useful contents
Today	The starting page: capture a thought, resume unfinished work, see completed results, and find anything awaiting me.
Work	The actual open document, spreadsheet, application, or investigation. The owner and assistant operate on the same current material.
Library	Browse, preview, compare, and organize material selected through the left-hand tree.

Preserve my place

Opening a document or working capability from Library brings it into Work. Returning to Today or Library preserves the open work, current selection, scroll position, and unfinished input where applicable. Returning to Work restores that place. Changing the visible destination does not itself pause or cancel an assignment; the existing work controls retain those roles.

Keep navigation and working context coherent

Give the owner clear routes to machine resources, files, settings, current work, conversation, and evidence. Group these functions where they help the task. Keep selection and focus consistent while letting the owner pin, collapse, or resize useful context. Fixed three-tab panels and equal columns are optional.

Give browsing controls a useful home

Make Icons, List, Columns, and Preview available through relevant view controls, menus, or context actions. Remember the chosen display style for each browsing location. Keep back, forward, up, and the location path together where they apply. These are functions to preserve; an Arrange menu or permanent workspace row is optional.

Put commands in the context, menu, or instrument where they are useful. Preserve actions, keyboard access, and focus when relocating them. Claude may refine labels and grouping while keeping capture, ongoing work, and the material collection understandable.

13 Color typography and attention

The BOSS brief and five hero references govern the current visual direction. Use charcoal and smoky gray surfaces, warm ivory text, copper-orange selections and actions, and restrained light-green signals. Mature manga ink and ASCII ornament give the shell character. Preserve room for real work and provide readable appearance alternatives.

Current shell palette

Shell color	Meaning and application
Charcoal and smoke	Layered backgrounds and quiet structure. Keep text regions clear and avoid decorative grain behind working content.
Ivory and gray	Readable primary text, supporting labels, tree branches, and table rules. Retain enough contrast at the owner's reading size.
Copper-orange	Current selection, available action, active destination, and focused command. Pair it with a clear label or focus shape.
Light green	A restrained local, ready, checked, or checkpoint signal when the record supports it. State exactly what was checked or saved.

This table now describes the BOSS shell palette. Native applications, images, charts, and the owner's documents keep their meaningful content colors. Pair color with words, icons, and focus. Tune exact shades for readable contrast and the owner's display preferences.

Terminal lettering gives the shell its voice

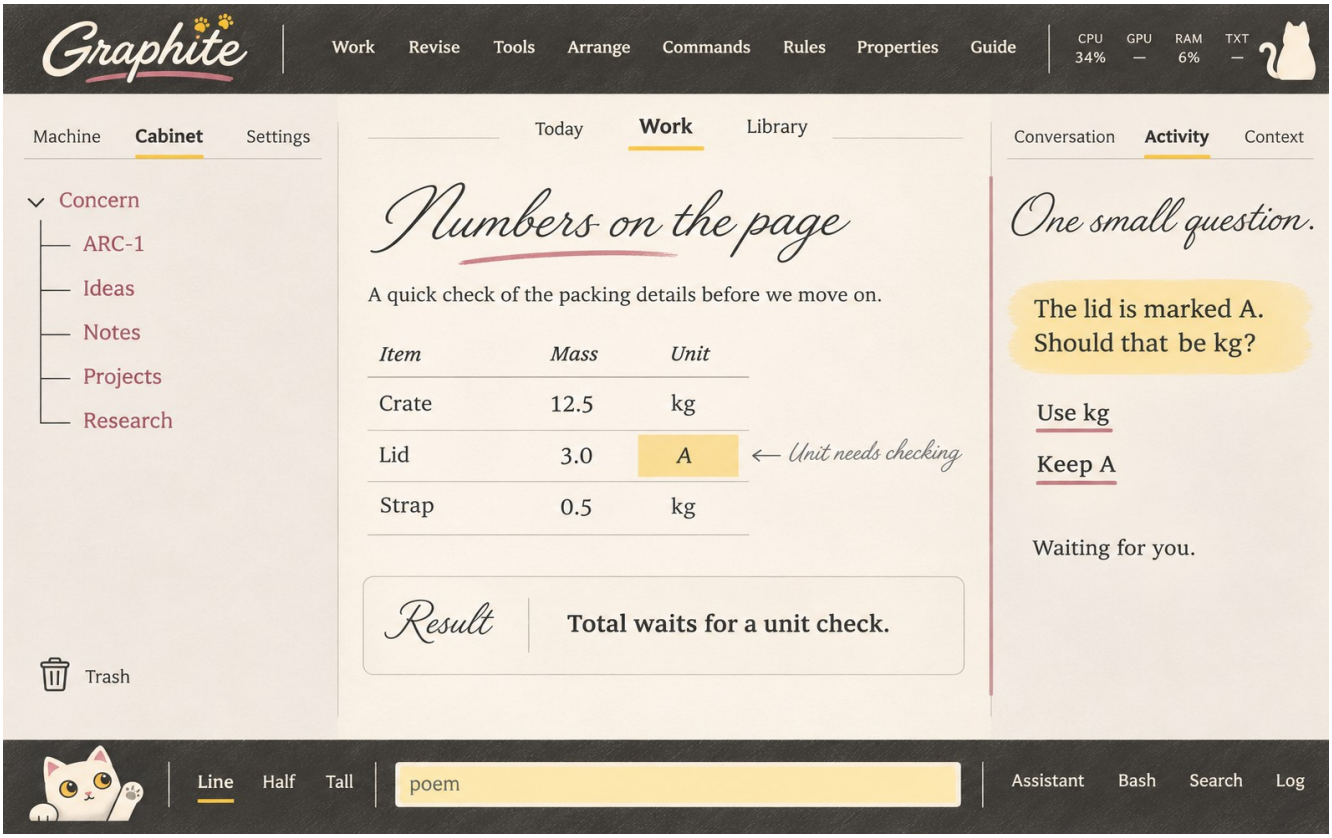
Use clear monospaced shell titles, compact terminal labels, and restrained ASCII motifs. Cursive is no longer the default interface voice. Keep decorative raster lettering to short headings where it remains readable; sustained text, quantities, units, commands, and controls use clear fonts at the owner's reading size. A document retains its own chosen typography.

Attention arrives as a useful note

An unresolved condition should point to the relevant value or object and a concise explanation. Offer useful next actions while keeping the rest of the work available. Color and optional character cues support explicit words and focus; they do not establish a required two-cat layout.

Give illustration and screentone deliberate spaces outside active text and controls. Use stable manga strips, sidebar art, or idle backgrounds; collapse decoration when the work needs room. Preserve shared commands and stable navigation. Claude may improve proportions, font choices, and artwork while carrying the BOSS visual character into real controls.

Earlier visual study



Historical visual study of a unit question. Preserve the question and source behavior; its cursive, yellow selection, and pink action styling are superseded by BOSS.

The lid is marked A among quantities recorded in kg. The example shows the intended response: keep the working page visible, identify the exact source of uncertainty, and ask a short question with useful responses. It illustrates interface behavior; these quantities are not ARC-1 measurements.

Display actual recorded states with access to the underlying evidence. Carry clear meanings across capabilities using words, position, and suitable visual cues. BOSS governs the current shell treatment; earlier colors remain historical studies.

The pictured footer is an earlier state. The current destination labels are Assistant, Terminal, and Log. Preserve search through explicit commands and relevant menus; simple local searches should work without a model turn.

14 Menus and contextual commands

Make the computer's capabilities understandable through contextual instruments, menus, keyboard routes, and commands that reach the same supported operations. Organize controls around the current work and discoverability. The older three groups of three head menus are optional material for the current direction.

Programs and System have clear roles

Programs opens a plain, understandable list of real applications and supported capabilities. Its affordance may be a label or an appropriate icon in a stable location. A fixed bottom-left cat is optional. A capability can open a workspace or expose a supported action.

System preserves real process and job controls, including the owned Stop action, work ordering, and relevant per-job niceness controls. Keep shutdown, restart, log out, and Return to Debian in their logical system location. Call the arcade entry Game. A fixed rear-cat control is optional.

Work order and operating-system niceness have different effects. Present the controls according to what they actually change. A slider should not imply that it directly reserves GPU time or makes an in-progress operation instantly preemptible unless the execution system supports that behavior.

Preserve the power of right click

Keep the Fluxbox-inspired dynamic context menu as a useful graphical route into relevant commands. Its contents should reflect the selected object, available capabilities, current artifact state, and the modifications that make sense there. Surface appropriate staC actions in a form the owner can understand.

Examples include acting on a selected quantity, sending a table into an available calculation, organizing a document into a sheet, comparing a source revision, or drafting correspondence about a current finding. Show an action only when its required context is available or explain the missing prerequisite naturally.

Keep interaction consistent

Preserve keyboard access, focus, Escape, opening directions, selection behavior, and useful hit targets. Commands reached through a menu, the shared typing strip, and an assistant tool should share the same artifact and authority rules. Help should explain a capability through a practical action rather than exposing internal module names.

15 Optional Graphite identity artwork

Retain earlier lettering, the pink underline, and paw-print artwork as historical assets. BOSS governs the default identity: terminal lettering, ASCII ornament, and mature inked illustration. Reuse older art only when it fits that direction or the owner selects it. The working desktop has no required upper-left wordmark or corner mascot.



Archived lettering asset with two paw prints. It is not the current default wordmark or a required desktop control.

Use identity artwork where it helps

Where historical lettering is intentionally reused, keep it legible and compatible with the selected appearance. Default shell headings follow the BOSS terminal typography. Keep functional space available for work and controls.

Choose bounds for the actual use and keep nearby controls readable. The desktop no longer reserves the upper-left and upper-right corners for the wordmark and cat.

Retain editable artwork and clear state semantics. Optional character animation should reflect actual work and should not displace controls or become the only explanation of system state.

16 Optional cat cues with clear meanings

The seven-pose cat artwork remains an optional historical cue vocabulary. The BOSS default does not require corner cats or a mascot. If the owner selects character cues, keep working, paused, sleeping, attention, and menu interaction distinguishable through words and controls as well as artwork.

Cue	Optional front view	Optional rear view
Idle	Sleeping face, closed eyes, relaxed paws.	Low curled sleeping mound, relaxed ears, tail wrapped around the body.
Working or wondering	Awake upward-looking eyes, slight head tilt, one paw under the chin.	Forward lean, tilted head, one cocked ear, loose S tail with a quiet sway.
Your turn	Friendly attentive eyes, head tilt, one paw raised.	Tall posture, sideways head tilt, asymmetric ears, question-hook tail.
Done	Brief pleased expression and happy eyes.	Brief upward stretch or perk, ears high, upright tail with a curled tip.
Paused	Patient half-lidded eyes, level head, both paws down.	Upright and still, level head, tail flat along the ledge.
Needs attention	Alert concerned expression with a subtle ear change.	Compact alert crouch, one ear outward, stiff diagonal tail.
Menu open	Happy and excited, both paws raised as if holding Programs above the head.	Leans toward the System menu edge with a reaching paw and a tightly curled tail.

Use the actual trigger

Idle means there is no owned work or unresolved attention request. Working means a real operation or model turn is active, including supported deterministic work or model loading. A queued job alone is not running. Your turn represents actual required input. Needs attention represents an actionable problem that prevents the intended work.

Show Done for an observed successful transition, never for cancellation or failure. A paused job retains a supported continuation point. A stopped or cancelled job remains stopped until the owner explicitly starts a permitted continuation or new attempt. Optional quiet poses may share artwork, but their accompanying text must preserve these different execution states.

17 Work cues and optional character interaction

Prefer one deterministic account of current work that feeds status text, activity views, and any optional character cues from existing jobs, questions, and worker lifecycle. Choosing a pose needs no model call. Preserve the event or assignment identity behind an attention cue.

Represent facts that can coexist

Active work, a request for input, an actionable problem, and menu interaction can occur together. An unresolved question can stay visible while another job runs. Choose a clear main pose and preserve the other facts in text or a compatible secondary motion. The owner should understand both the need for input and the continuing work.

For example, a raised-paw face can retain the input request while a hooked tail sways gently for another active job. The status can explain that input is needed and other work continues. Keep this representation tied to current records rather than maintaining an independent decorative job state.

Menus are an immediate interaction

If optional character artwork supplies a menu overlay, adapt it to the actual Programs or System affordance. Opening a menu changes presentation only; it does not start, pause, or complete work. When the menu closes, restore the cue from the current recorded state. No second corner character is required.

Hover, keyboard focus, menu highlighting, and work status should cooperate. Opening a menu does not pause a job, create an input request, or turn a completed job back into active work. The paws suggest holding or reaching toward the menu without moving its anchor or interfering with selection.

Keep motion quiet and inexpensive

Use the existing Qt animation machinery with small movements. Tail swaying should reflect active work. Idle settles into sleeping; new work or attention wakes it. Stop animation when a view is hidden or destroyed. Respond immediately to the existing reduced-motion preference with distinct still poses and current text.

Keep accessible action names such as Open programs and System menu, and include meaningful status in descriptions or tooltips. Update those words on state changes, not every animation frame. The animated cue should remain a useful companion to ordinary work reporting.

18 Optional pose reference for native artwork

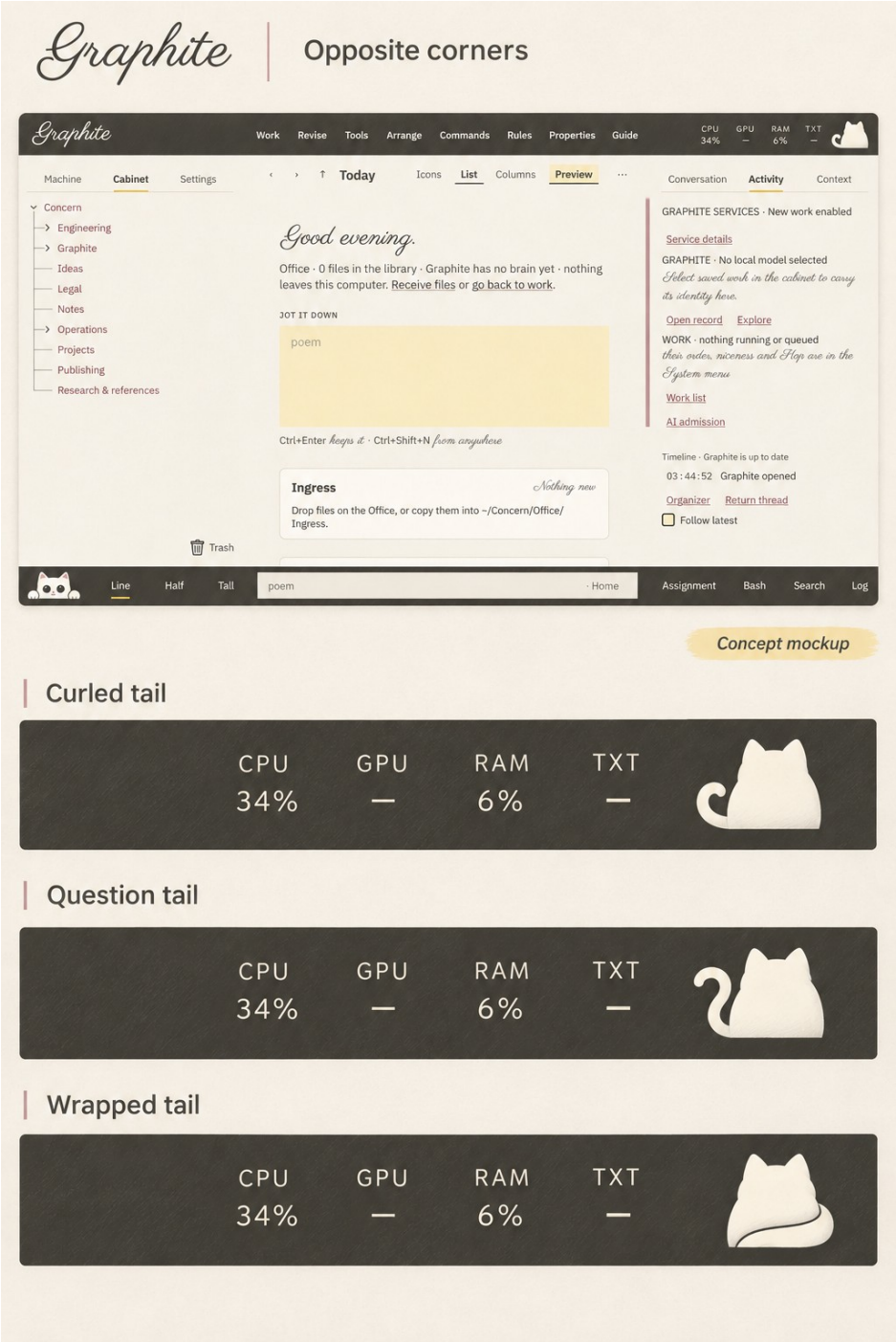
This optional board preserves expression and silhouette studies. Use the cue definitions if a character is adopted, and fit the artwork to its actual control. Differences between older lettering treatments are asset history; they do not establish a required current wordmark or location.



Concept reference for seven poses. Menu open is an interaction overlay. Working and paused remain distinct when motion is reduced.

19 Archived placement reference

This is an archived placement study with a front cat below, a rear cat above, and an earlier three-column composition. Preserve it as optional reference material. Current requirements concern discoverable Programs and System access, readable work cues, and the objects and instruments desktop; the pictured corner positions and wordmark placement are not binding.



Archived placement concept. If the rear silhouette is used, it remains a tidy back, ears, tail, and useful paw gesture without anatomical marks. The image is not a required screen arrangement.

20 Optional native artwork and layout behavior

If the optional cats or lettering are used, maintain editable artwork and relevant notices. Scalable Qt paths, suitable licensed lettering, or another compatible construction may fit. The reference PNGs guide character and expression; they are not finished production sprites or a requirement to implement a character system.

Distinct at real control sizes

Give the rear cat visibly different body height, lean, head angle, ear position, and tail shape across cues. A sleeping body, an upright paused body, an active forward lean, and a successful perk should remain distinguishable without relying on animation. The front view should convey its cue through silhouette and expression as well as color.

If character cues are used, choose gestures that fit the real control and menu direction. Keep Programs and System discoverable through labels and normal input regardless of whether a mascot is present.

Geometry does not change with a pose

Where animation is used, allocate bounds for its largest pose. Keep menu anchors, nearby controls, and resource readings stable and unobscured. Do not require mirrored panels or fixed band heights to achieve this.

Fit any identity artwork using its actual ink extents. Keep optional cues readable at the chosen scale and apply the owner's reading and motion preferences. Two opposite-corner silhouettes are optional.

Preserve existing control behavior

Retain menu, focus, keyboard behavior, and real job controls while changing their presentation. Existing marks can remain useful for pointer, startup, window, or identity assets where appropriate. Current desktop placement follows sections 70 to 77.

Use relevant existing presentation checks for cue transitions, mixed work and input, menu overlays, reduced motion, and small-size fit. Actual captures should label simulated cue data where it was used. The visual references are not evidence of implemented or owner-host behavior.

21 Desktop delivery and the ThinkPad profile

Deliver Graphite as the desktop session on a maintained Debian base, with the owner's ThinkPad E16 Gen 2 as the first hardware profile. The reported configuration is Ryzen, integrated Radeon, and 64 GB of shared system memory; detect the actual unit and effective limits. Keep native application windows, normal input, clipboard, file selection, and a Debian terminal usable within the overall environment.

Know the machine being configured

Record the CPU and GPU identifiers, usable memory and its configuration, operating-system and kernel versions, Mesa and Vulkan information, selected runtime, and model identity. Lenovo's published E16 Gen 2 AMD specification includes Ryzen 7 7735HS and 7735U configurations paired with Radeon 680M [R06]. Use that family information as context for detection rather than assuming the suffix or processor from an incomplete description.

Expose useful effective information under Machine: the loaded model, actual device and backend, and supported offload information reported by the worker. Keep user-facing labels truthful when a CPU path is active, a Vulkan path is unavailable, or a setting cannot apply to the selected model.

Specialize the computation boundary

The owner's goal is to make the existing program use this machine well. Place hardware-specific choices in the runtime and configuration boundary where possible. Preserve the general evidence, operation, and artifact contracts so a future computer can use an appropriate profile without losing project knowledge.

Python and Qt can continue serving the application where they already fit. Native kernels, established runtimes, or targeted lower-level code can perform the demanding computation. Choose implementation changes for the behavior and bottleneck they address. A broad rewrite of working application code requires its own concrete benefit.

Useful sustained operation

Keep the desk responsive during local inference and longer work. Account for available memory, bandwidth, power and thermal conditions when the host exposes them. Avoid a configuration that makes ordinary document work unusable simply because a larger model can be allocated.

Keep the detected profile, dependencies, and reasons for material limits with the configuration. The primary complete-system delivery is a bootable USB image with a supported installation and recovery route. Explain live-session persistence and save locations, preserve user work across upgrades, and distinguish application updates from disk installation. Hardware-dependent observations retain the identity of the machine on which they were obtained.

22 Ryzen Radeon and shared memory

Use the two processors as complementary resources. Ryzen arranges data and runs explicit methods. Radeon performs supported neural prompt processing and generation through the selected Vulkan backend. The runtime may place auxiliary operations where its implementation requires them; keep the intended division useful and technically compatible.

Capacity belongs to the whole system

The installed 64 GB is one physical system pool shared by the CPU, integrated GPU, operating system, and other work. Account for model weights, attention state, runtime buffers, CPU evidence and tools, and the desktop together. Avoid adding overlapping host and GPU accounting as if it represented separate physical memory.

An integrated GPU can expose memory beyond its firmware-reserved graphics allocation, subject to the driver, runtime, and heap limits. Choose placement using actual supported allocations and current headroom. Do not equate installed capacity with a guarantee that a particular Vulkan allocation or model configuration will succeed.

Bandwidth and copies matter

CPU preparation and GPU inference can overlap usefully when they leave enough resource headroom. A short indexed lookup may coexist comfortably with inference. A large scan, extraction, or conversion may compete for the same memory bandwidth and should yield when that helps the active desk.

Inspect avoidable duplicate weight buffers, repeated uploads, format conversions, and unnecessary copying at the inference boundary. Reuse compatible buffers and prepared state where the backend supports it. Shared physical RAM still requires the Vulkan implementation's allocation, synchronization, and ownership rules.

A practical budget

Use available host and device information as part of a coordinated budget. Vulkan's memory-budget extension provides implementation-dependent heap usage and budget estimates that can change with system activity; treat them as guidance for the current process [R04]. Keep a useful allowance for the active workspace and tool workers.

The reviewed CPU worker uses a virtual-address-space limit. GPU drivers can reserve virtual address ranges differently from their physical use, so adapt that protection to the chosen backend. Preserve bounded resource use through an appropriate mechanism, and leave an oversized request ready for a suitable model or configuration decision.

23 Traffic cop coordination

Implement traffic-cop coordination through Graphite's existing work ownership, application Kernel service, agenda, agent loop, and model manager. Here Kernel means Graphite's coordination machinery, not a replacement Linux kernel. Choose among explicit computation, evidence retrieval, model interpretation, waiting for information, and permitted preparation according to the assignment and resource budget.

Let the owner's work lead

A foreground request takes precedence over queued preparation. A sustained investigation can use a larger budget when the owner has assigned it, while still yielding at supported operation or model boundaries. Distinguish a requested priority from the actual preemption facilities exposed by the worker.

Coordinate the shared memory traffic as well as CPU and GPU occupancy. Combine rapid changes to the active selection, schedule a large source update around active inference, and discard preparation that is no longer useful. Extra activity should have an identifiable benefit to the work.

Prepare only useful safe candidates

Look ahead at cheap, permitted reads when their concrete inputs are already known. Store the result with the tool and source identities. Consume it only when the selected action and current inputs match. Writes and external effects go through their normal authorized operation when that action is actually selected.

Research on overlapping predicted tool execution with model generation offers a candidate design direction [R11]. Its serving results do not establish a speedup on this laptop. Use the relevant established idea only where it fits Graphite's operations, resource budget, and applicability checks.

Interruption has a real meaning

Pause preserves progress at an appropriate boundary. Stop cancels the owned operation and pending preparation. A user cancellation remains stopped until an explicit owner action resumes supported work or starts a new attempt. A crash or execution failure has a separate recovery path and should not silently reverse the owner's cancellation.

If memory pressure releases a prompt cache, reconstruct what is needed from durable evidence. If a worker fails, identify the unfinished step and reconcile any uncertain result. Keep these events distinct in the activity record. A lower-level scheduling decision should not lose findings, invent completion, or conceal an unresolved question.

24 Intelligent data arrangement before a model turn

Assemble a useful working picture from permitted material and explicit methods before each relevant decision. Model interpretation can establish which material or calculation matters. Let CPU preparation, interpretation, execution, and observation alternate as the assignment develops. Reuse typed records and mathematical operations while retaining the conditions that determine meaning.

Information carried	Meaning to preserve
Goal and active work	The intended outcome, operation, current question, selected artifact, applicable revision, and a basis for assessing completion.
Source identities	References to the passages, records, files, and versions actually used.
Facts and conditions	Values, units, roles, ranges, applicability, extraction uncertainty, source context, and measured or assumed status.
Completed computation	Actual results, method identity, input bindings, and the limits of the calculation.
Open conditions	Missing information, conflicting sources, alternatives, and unresolved assumptions.
Available actions	Relevant supported operations, concrete inputs where known, and permitted effects.
Budget and context	Enough context for the decision, room for a useful answer, and the assignment's limits.

A semantic agreement

Preserve these meanings through the current records rather than a mandated format. Establish the intended operation and roles: incoming or outgoing, original or replacement, item or subtotal, included or excluded, sequential or parallel. The model can propose an interpretation; deterministic checks assess supported expressions, units, references, and constraints. Exact arithmetic alone does not establish task correctness.

Use calculations, typed filters, and dependency relationships where they answer an explicit question. A relevance ranking can narrow candidate sources. Keep its meaning distinct from the model's expressed certainty and from a measured probability. Retain enough source detail to investigate an exception or ambiguous claim.

Use observed results to settle conditions, expose unsuitable methods, and identify the next useful question. Pursue that question for its effect on the decision, using available sources and tools. Retain unresolved assumptions and retrieve more context when a summary cannot settle the interpretation.

25 A workspace that comes prepared

When the owner settles on a document or an assignment, let the system prepare the relevant evidence during a useful quiet interval. The aim is to have a compact, current working picture ready when the next question arrives, without making preparation compete with the foreground work.

Keep three different resources understandable

Source indexing makes records findable. Model residency keeps the loaded weights available. Prepared model context retains supported temporary inference state for a matching prompt prefix. These are separate choices with separate resource costs, even when the owner experiences them as one attentive workspace.

Let Ryzen assemble current evidence and known computations. If the worker is free and memory permits, a supported runtime can process the stable prefix before the question. Use a matching effective prompt format when the actual question is appended so available prefix state can be reused.

Use supported runtime behavior

The native llama.cpp completion interface documents n_predict set to 0 for evaluating a prompt without generating text. It also documents cache_prompt for reusing a common prefix. These are candidate facilities in the chosen pinned worker; a native completion option should not be assumed to exist unchanged in every compatible chat interface [R01].

A shared prefix can reduce repeated prompt processing. It does not by itself make the generation of new tokens faster. Treat prompt preparation and generation as different performance problems. vLLM's prefix-cache documentation makes that distinction explicitly [R09].

Prepare within a useful bound

Start with a controlled preparation for the active work, coalesce rapid selection changes, and let foreground requests supersede queued preparation. A cold request still needs an ordinary path through current evidence. It can use valid partial preparation or arrange what it needs directly.

Keep applicability and eviction separate. A valid cache may be discarded to release memory. A stale cache must not be reused because it remains resident. Explain a useful outcome through the existing activity reporting, such as a relevant earlier investigation found or a current calculation available. Cache management itself should remain under the hood.

26 Identity revision and cache validity

Prepared state belongs to a particular model, runtime, prompt construction, source revision set, and permitted project context. Use that identity when consuming a prepared result and accepting a returned finding or action. An unchanged screen is not sufficient evidence that all of those inputs remain current.

Invalidate what changed

Scope reuse to the loaded model identity, relevant runner and tokenization settings, chat template, prompt mode, owner and project access, and evidence revisions. Associate prepared state with the actual worker and session or slot holding it. Include any adapter or other configuration that affects the effective computation. Changing these can invalidate derived state.

Recheck current applicability at the boundary where the prepared result is consumed. An artifact can change while inference is running. Preserve revision binding on the proposed operation and retain the current rules that decide whether the action can still apply or needs fresh evidence.

Keep a trustworthy model identity efficiently

The reviewed reply path reads and hashes model files on every turn. Examine a lifecycle that avoids unnecessary repeated full-file work while retaining a trustworthy identity for the bytes actually loaded. Full digest verification belongs at registration, activation, replacement, and where the bytes may have changed.

A verified owned snapshot is one possible mechanism. Another compatible approach may fit the existing runtime better. Preserve change detection and race protections. Describe the guarantees of the faster path, associate resident weights with their verified identity, and revoke affected prepared state when that identity changes.

Make temporary state disposable

Memory pressure, worker replacement, or a crash can remove cache state. The assignment's evidence, actual tool results, and open questions remain durable. Reconstruct the prompt from those records when needed. A worker cache is an acceleration of known work, not the only surviving account of what Graphite has learned.

Apply consistent policy across ordinary replies, streaming replies, and protected model requests. Preserve the authority boundaries of those paths while giving them compatible handling of prepared evidence, revision checks, cancellation, and model identity. The owner should receive coherent behavior whichever presentation is active.

27 Reuse mature inference methods

Use established implementations for a concrete need. Choose methods supported by the selected model, Radeon backend, runtime release, and maintenance budget. The projects below are sources of compatible code or established design; their presence here does not require importing each complete serving stack.

Method or source	Purpose and compatibility question
llama.cpp Vulkan	Supported neural kernels, layer placement, attention behavior, and model formats [R02, R07].
Precision and microbatch controls	Shape weight, cache, and temporary memory costs using facilities the chosen backend exposes.
Prefix reuse	Avoid repeated work for a matching effective token prefix [R01, R09].
Speculative generation	Consider compatible draftless n-gram prediction or model-specific prediction heads [R08].
vLLM cache organization	Consider compatible paged allocation and prefix-sharing design for attention state [R09].
SGLang session cache	Consider session-aware retention and eviction under memory pressure [R10].
MLC LLM compilation	An alternative requiring converted weights and a compiled model library, including a Vulkan target [R13].
Expert placement for MoE	Use supported CPU expert placement or GPU expert caching for an appropriate mixture-of-experts model.

Fit the actual backend

GPU kernels depend on their API, data layout, types, and supported architecture. Prefer direct reuse of a compatible implementation. Where adaptation is appropriate, carry over the method and its relevant invariants. A CUDA implementation does not automatically become a Vulkan implementation because both target a GPU.

Retain versions and upstream notices with reused code. Keep effective capabilities tied to the pinned worker. If an alternate runtime needs different model artifacts, preserve the existing model-manager contract and explain how those artifacts are obtained and identified.

28 Choose the model around the work

A local model is an instrument in the larger workspace. Preserve registration, selection, replacement, and inspectable identity. A model change should leave the accumulated project knowledge available and should invalidate temporary inference state derived from the previous model.

Allocate context deliberately

Use the selected tokenizer and effective prompt to account for context space. Leave room for the answer or structured action the task needs. A short constrained choice can use a modest turn. A difficult interpretation may need more evidence and a longer answer. A configured maximum should not make every request consume the maximum.

The reviewed path uses enable_thinking set to false. Treat any change to this model-specific option deliberately with its context, generation, time, and cancellation implications. Preserve useful interpretation rather than forcing every question into a yes-or-no response.

Treat different memory choices separately

Weight quantization, attention-cache precision, context size, prompt microbatching, and model selection affect different parts of the work. Use supported Flash Attention or placement choices where they fit the actual backend. Record the effective settings so the owner can understand which configuration produced a finding.

A larger model can occupy more of the shared pool and change what tools or contexts can remain resident. A second draft model also consumes memory and work. A mixture-of-experts model requires a workable strategy for the full weight set and access pattern even when an individual token activates only some experts.

Keep behavior meaningful on the machine

Choose a model for the task and available resources. Do not equate a parameter count, a quantization label, or the ability to allocate weights with reliable task completion. Explicit methods and current evidence can help a small model, while its interpretation limits still need to be represented honestly.

If the GPU path fails, use a compatible CPU path when the same model and job fit the current budget. Explain the active backend and the reason for fallback. Preserve completed work and the owner's cancellation intent. A resource or model decision should remain an understandable request when an automatic continuation is unsuitable.

29 Vulkan runtime and dependency delivery

Upgrade the existing inference path through Graphite's ordinary installation and update mechanisms. Deliver a pinned Vulkan-capable worker or a compatible build with its runtime dependencies. Keep the CPU path usable, and preserve current GGUF registration and task entry behavior.

Use the supported Linux graphics stack

Vulkan is a cross-vendor GPU API. RADV is Mesa's Vulkan driver for modern AMD graphics and works with the Linux amdgpu kernel driver [R03]. Use the host's compatible kernel support, AMD graphics firmware where needed, the libvulkan1 loader, and mesa-vulkan-drivers. Debian's AMD firmware package is firmware-amd-graphics [R05].

Include vulkan-tools when useful for device information and diagnostics. Keep build-time headers, SDK components, and shader tools in the worker build process rather than treating every development dependency as a mandatory owner runtime dependency. Match the delivered components to Debian 13 and the actual host configuration.

Pin what is delivered

Record the worker version, archive digests, dependency versions, supported model and backend capabilities, and third-party notices. The upstream llama.cpp build documentation currently uses CMake and a GGML_VULKAN option for Vulkan builds [R02]. Use the interfaces of the selected pinned release when integrating or packaging it.

Choose reliable packages or a compatible redistribution mechanism under the existing installation flow. Public availability does not make every driver or dependency freely redistributable under identical terms; carry the applicable notices and distribution requirements with what is actually bundled.

Explain the effective state

The Machine view should show the device and backend actually used. A Vulkan build without an active Radeon worker is not GPU execution. Report a missing dependency or incompatible feature clearly and continue through a suitable supported path when possible.

The bootable Graphite desktop is the primary complete-system target, alongside a usable application installation and upgrade route for ongoing development. Build the image from a maintained Debian base and suitable redistributable components, with dependency and model-acquisition instructions where separate downloads are needed. Inference improvements can land through the application route while the image is completed; the image is part of the destination rather than a discarded or blocking prerequisite.

30 The reviewed inference starting point

The supplied inference brief records observations from a reviewed Graphite 0.1 release. Its five release volumes and 1,101 payload files were reported to match their manifests. The following notes locate that baseline. Claude should inspect the current checkout before making changes, since these paths and defaults may have evolved.

Existing area	Recorded behavior in Graphite 0.1
runtimes/llama.cpp/b11669/ runtime.json workbench/brain_runtime.py	Bundled Linux runtime pinned to a CPU build at b11669.
workbench/local_brain.py configuration and start_server	Launch uses -ngl 0; default context is 2,048 tokens; at most four threads; automatic model budget capped at 32,768 MiB.
workbench/local_brain.py completion and run_reply	Tokenizes the prompt, checks context space, and explicitly disables prompt caching.
workbench/slate_stream.py workbench/worker_tasks.py	Streaming and protected model requests also disable prompt caching.
workbench/brain_worker.py	Native Linux worker applies RLIMIT_AS, a virtual-address-space limit.
workbench/local_brain.py recheck sdk/cli/arc1cli/gguf_inspect.py	Reply rechecking reads and hashes model contents; the inspector supplies identity protections.
workbench/agent_loop.py workbench/agent_profiles.py	Records real operations and can ask the model to choose among candidate actions.

Integrate through the existing contracts

Use these notes to find the current boundaries for a Vulkan worker, task-specific configuration, valid prefix reuse, and a more efficient identity lifecycle. Preserve the job, operation, protected-request, and artifact rules that already govern real work.

These observations are code and release-review context. They do not establish owner-host GPU operation, the exact ThinkPad configuration, or the performance of a model on the owner's machine. Keep that distinction when reporting the upgrade.

31 Existing records and presentation integration

Build on the current records and interface foundations. The supplied briefs identify the following Graphite 0.1 entry points. Locate their current equivalents and inspect caller contracts rather than copying older code into a newer checkout.

Existing entry point	Relevant foundation
References and knowledge services	Source identities, project access, retrieval, and current evidence.
math_service and math_backend	Explicit computations and mathematical methods.
Investigations and conditions	Durable assignment work, findings, and unresolved applicability.
reason_graph and typed_artifacts	Relationships, revision bindings, and structured outputs.
checked_methods and agenda	Retained useful procedures and owned scheduled work.
workbench/graphite_brand.py	CatPoses, POSES, paint_cat, and Cat; Qt paths, pose blending, hover, open, sleep, and reduced motion.
workbench/hatch_qt.py workbench/typebar_qt.py	HatchButton owns the front cat and Programs interaction; TypeBar places it.
workbench/desktop_header_qt.py workbench/system_menu_qt.py	SystemButton and SystemMenu own the head control and its real menu behavior.
workbench/desk_qt.py	Shared band geometry, script_word_px, word_stroke, and ink measurements.
workbench/agent_qt.py workbench/agent_loop.py workbench/agenda.py	Existing assignment and worker status, including RUNNING, WAITING, PAUSED, DONE, STOPPED, FAILED, and needs_you.
workbench/reading_profile.py workbench/settings_registry.py	Reading and settings foundation, including reading.reduce_motion.

Extend the model settings registry and shared logic where suitable. Keep one coherent account of work across the contextual inspector, job controls, and any optional character cues. New capabilities should contribute through the corresponding records and operation boundaries. Locate the current equivalents of historical code paths before changing them.

32 Lasting methods and extended work

Completed work should leave the next assignment better informed. Preserve useful methods, corrected evidence, and the circumstances in which an approach worked or failed. Use the current investigation and method stores so a fresh model session can retrieve that knowledge.

Keep helpers as usable tools

When Graphite writes a small helper to solve an assignment, retain it if it has reusable value. Record its purpose, inputs, expected outputs, dependencies, applicability conditions, and observed result. Give the Tool Atlas enough information to discover it later. Keep its execution within the current supported operation and authority mechanisms.

A saved calculation should retain units, assumptions, input bindings, and the method that produced it. When a later request meets those conditions, reuse the method through its registered operation. If an input is missing or the conditions no longer apply, expose the gap before treating the result as useful evidence.

Remember unsuccessful approaches precisely

An attempted method can reveal a bad assumption, unsupported format, ambiguous source, or tool limitation. Keep the actual reason and affected conditions. A later model turn can use that information to choose a better next action. A failure in one context should not become an unconditional rule against a method that works under different conditions.

Work while the owner is away

Support sustained assignments within the owner's time, resource, and capability scope. Continue by pursuing useful open conditions, recording the actual results, and arranging the next relevant question. If evidence or authority is missing, complete the independent work and leave a specific question or requirement.

The completed system should support owner-authorised work continuing with the Graphite presentation window closed through a real persistent runner. Keep assignments, checkpoints, and service controls reconnectable when the workspace opens again. Ending a graphical session or powering off has different effects, described in System. If a development build can only pause when its window closes, report that as the current limitation while completing the required background behavior.

Use workers where they help

Separable research, extraction, calculation, or review can use cooperating workers when the resource budget and actual tools support it. Keep one shared assignment record, clear artifact ownership, and a place to reconcile findings. Additional workers should earn their cost through useful work rather than becoming a required collection of personalities or always-resident models.

33 A complete engineering investigation

Imagine I provide the authorized ARC-1 collection, described by the owner as thirty-three documents and roughly seventeen hundred pages, and ask whether a proposed design can be built. Verify the actual intake inventory and revisions. Graphite should organize the material, retrieve relevant passages and drawings, calculate from applicable values, and investigate unresolved requirements. This describes intended behavior rather than a completed review of that collection.

Let tools establish what they can

A datasheet can become a typed worksheet with page references. The CPU checks units, performs available calculations, and follows dependencies. A compiler or engineering tool can provide an observed build, model, or analysis result when that capability is present. Each result enters the durable record with its inputs and conditions.

The next question may concern a conflicting variant, an applicability condition, or the observation most likely to change the decision. Select it from the actual evidence gap and investigate through local material, suitable tools, or permitted sources. Ask the owner when their judgment, access, or a physical observation is needed. Keep model proposals bound to current artifacts and authority.

Keep changing evidence coherent

If I correct an extracted value or add a measurement, Graphite updates affected findings and prepares the next question from the revised evidence. It keeps the earlier attempt and its source versions. A draft report or unsent technical question using the old result should be reviewed accordingly.

A useful investigation can prepare the related paperwork: an explanation of the finding, a concise question to a component supplier, an agenda for a design review, and a meeting arrangement using the relevant availability. Those actions draw on the same current understanding and the owner's granted scope.

Readiness needs a defined basis

The owner's ambition of a 95 percent chance of first-build success is a meaningful target only with a defined outcome and a defensible basis for the probability. Preserve checked requirements, unresolved conditions, simulations, measurements, and model interpretation as distinct evidence. Do not manufacture the probability from a relevance score or the model's confidence language.

A result may be that specified checks pass under stated conditions and that other evidence is still required. That finding can still advance the project substantially. Graphite should leave the owner able to inspect the basis, understand the remaining uncertainty, and decide or delegate the next useful action.

34 The same depth for other assignments

ARC-1 exercises a demanding combination of sources, calculations, tools, and decisions. The same foundation should handle other work introduced through the owner's intent and available material. Avoid designing the general assistant around a single project's names or predetermined answers.

Writing and document work

I can bring notes, email chains, PDFs, images, and source material, explain the intended audience, and ask for a useful document. Graphite should retrieve relevant evidence, organize it, edit the actual artifact, and prepare a readable printable result. A letter can quote selected correspondence and an applicable published policy, with supporting references retained and space for a handwritten signature when requested.

Software and tool work

I can describe an issue and provide the code or logs. Graphite can locate relevant material, reproduce the condition when the available environment supports it, use the installed compiler or tools, make a suitable change, and record the observed result. A useful helper can become a retained method for later work.

Research and comparison

I can ask a question spanning local files and new web material. Graphite retrieves evidence, organizes a comparison, and keeps disagreements visible. This can include finding images by a stated characteristic or investigating authorized network observations with suitable tools. Preserve the basis of each finding; an unusual observation and a supported diagnosis have different meanings. Revisit affected results after a source revision.

Everyday coordination

I can ask what I need for a meeting, whether a message needs a reply, where a relevant document went, or which commitments depend on unfinished work. Graphite uses the actual records and available tools, prepares a useful response, and performs authorized changes. The owner should not have to repeat the whole project to obtain each small result.

Adjust effort to the assignment

A simple question should not automatically trigger an extended investigation. A difficult question should be allowed enough evidence and iteration to become useful. Keep the desired depth, relevant tools, current state, and practical budget attached to the job.

These examples describe reusable behavior. New subjects and capabilities can join it through the same source, method, artifact, and operation contracts. The assistant's knowledge of what it can do should grow as the actual installed toolbox grows.

35 Local ownership and connected capabilities

Keep model inference and accumulated project knowledge local by default. Documents, calculations, preferences, findings, and methods remain useful offline. The Graphite core is intended to be free of charge, with no mandatory subscription or cloud-model account. The owner can connect chosen web, mail, calendar, or other services for assigned work; their accounts, costs, and data flows remain distinct from local inference.

Local processing and external effects

A local model can interpret an imported message and prepare a reply offline. Sending that reply uses the chosen mail service. A local calendar can support planning offline; shared availability and invitations depend on the connected calendar. A web source is external even when its content is subsequently indexed and reasoned about locally.

Make those distinctions visible in the relevant state and records. A missing connection should not erase the value of local preparation. A connected capability should not quietly replace the local inference architecture or send the whole project to an external model. Use only the data needed for the authorized operation.

Use the authority already given

Apply existing assignments and standing grants to the relevant account, project, and action. Routine authorised work needs no repeated permission request. Manual Run executes a constructed action; delegated assignments can execute supported steps within scope. A preview explains an effect and does not itself create another approval requirement. Ask only for missing intent, access, or authority.

When a consequential action needs a decision, prepare it concretely so the owner can inspect the intended effect. When the relevant authority exists, act and record the observed outcome. Keep preparation, execution, acknowledgement, and uncertainty distinguishable, especially across a service interruption.

Keep the owner in control

Pause, Stop, priority, and inspection must continue to affect the real work as capabilities grow. The owner should be able to understand which job owns a process, which records informed a finding, and which operation caused an external change. Preserve the current protections and recovery mechanisms while making the assistant useful and independent within its scope.

Knowledge and useful methods live in Graphite's records rather than depending on one model's context or weights. Keep them available when a model is replaced, a cache is released, or a worker restarts. That continuity is central to the owner's practical control of the system.

The lasting product is the owner's organised body of work: sources, artifacts, methods, decisions, assumptions, and open conditions. Internal databases can coordinate it; supported exports provide ordinary readable files and relationships for another person to inspect and continue without Graphite where practical.

36 Develop toward useful completed work

Claude, use this specification to connect and complete the current program. Inspect the actual implementation, identify the gaps that prevent the intended behavior, and choose the development order. Preserve useful work, improve what needs improvement, and use your creative judgment to add capabilities that advance the goal.

Make implementation choices concrete

Reuse mature tools and compatible runtime features. Adapt existing code when it already provides the appropriate boundary. Build missing helpers or integrations when they have a clear purpose. Material departures from a suggested technique should be explained through the owner behavior they improve and the costs they introduce.

Keep the code and artifact records coherent across the inference upgrade, capability growth, and interface work. The owner should receive a program that does useful tasks through the visible workspace, with its deeper machinery inspectable when needed. Resume the current construction goals with these stronger connections rather than treating this document as a separate product rewrite.

Verify the affected behavior proportionately

Use existing checks and focused verification for changed behavior. Select relevant tasks and exceptions from this brief instead of automatically rerunning all thirty-five scenarios or full suites after every increment. The historical stock-versus-total failure illustrates procedure applicability: the intended remaining-stock answer was 9 kg, not the computed 15 kg total. Assess the requested outcome and saved artifact; broaden checking when a change or observed failure warrants it.

Use current operation, artifact, service, and interface checks for the changed boundary. Presentation checks should cover targets, revisions, edits, handoffs, focus, and owned Stop. Check optional animations only when changed. Historical test paths are lookup hints, not targets for test counts or new presentation campaigns.

Report what the owner can now do

Keep the report concise: useful behavior delivered, concrete results observed, material choices, current limitations, and anything that depends on the ThinkPad, its model, or a connected account. Include actual interface captures where they help explain completed work. A large test count or a visual mockup alone does not establish an implemented end-to-end capability.

The destination is a usable computer: I can work directly, delegate an edit or sustained investigation, and return to inspectable progress. Its accumulated knowledge should support technical work and the correspondence and arrangements around it.

37 The desk as a working composition

I want to recognise my place immediately, whether writing a sentence or returning to a difficult investigation. Let related work share a coherent surface and give the active artifact the space it needs. Keep navigation, machine information, and assistant activity available through stable controls without forcing permanent equal side columns.

Preserve the frame

Use aligned controls, readable hierarchy, consistent selection, and balanced spacing. Let the owner pin or reveal useful context and change pane sizes. Keep stable navigation and command destinations while allowing an object-centred composition to use the display well. Equal columns and matching shaded head and foot are optional.

Where optional artwork or animation is used, keep it within stable bounds so it does not move controls or cover useful work. Preserve readable resource and model information.

Let the active work set the density

A blank page can be spacious. A spreadsheet needs more rows and columns. A design application may need nearly all of the working surface. Adapt the center to that material while preserving the surrounding anchors. Secondary tool controls can appear with the active capability and recede when it changes. Avoid leaving abandoned toolbars behind.

When space becomes tight, preserve a useful artifact viewport, the shared input, and a clear route to each side panel. Retain the owner's chosen widths and reading preferences where compatible with the available display. Explain any temporary collapse through a visible affordance and restore the previous arrangement when space returns.

Make movement predictable

Changing a tab should preserve the meaningful place within it. A scroll position, tree expansion, unfinished request, or selected object belongs to that destination. A new result should not move the pointer target or pull the owner away from current writing. Use a quiet cue and a direct route to the result instead.

38 Terminal lettering and readable work

Graphite's current interface voice is clear terminal lettering with restrained ASCII detail and mature manga ink. Preserve grace through spacing, hierarchy, and carefully placed art. The earlier cursive treatment remains historical artwork. Use real text for headings and controls so the shell scales and remains accessible.

Give each kind of text its role

Use monospaced headings and technical labels with distinguishable characters. A short display title may have a raster character; sustained prose needs a readable text face. Quantities, units, commands, paths, and identifiers must stay unambiguous. The owner's document fonts and native application typography remain part of their actual content.

Use a small, consistent hierarchy of titles, section labels, working text, and supporting notes. Establish importance through size, weight, placement, and spacing before introducing more color. Keep supporting notes readable at the owner's preferred scale. Technical detail can expand into an appropriate view when it needs room.

Preserve the voice at real size

Judge lettering at its actual size. Keep headings modest enough to leave room for content and preserve clear focus. ASCII illustrations occupy deliberate decorative regions; command text remains selectable text. Provide a plain readable alternative when ornamental lettering is difficult to read.

Use compatible licensed fonts or maintainable native artwork. Preserve their notices and editable sources. Provide a readable fallback where a preferred face is unavailable, retaining the same hierarchy and approximate proportions. Avoid a startup layout that changes dramatically when a font finishes loading.

Readability belongs to the owner

Respect the existing reading and reduced-motion preferences throughout menus, tables, side panels, and dialogs. Text scaling should improve legibility without hiding actions or cutting off labels. Preserve keyboard focus and the owner's place during a reading adjustment. The personal visual character should survive these practical accommodations.

39 Color meanings across changing states

Apply the BOSS shell colors consistently: charcoal and smoky gray structure, warm ivory working text, copper-orange actions and selection, and restrained light-green signals. Content and native applications keep useful colors. Preserve contrast and pair state with explicit words and position.

Keep attention local

When a value needs checking, identify that value and the related condition. Preserve the rest of the page as working material. Show the active writing place and current selection clearly using consistent focus treatment and labels.

A running job can have a restrained activity cue accompanied by its actual state. Reserve stronger emphasis for the specific input or condition needing attention. Several concurrent jobs should leave the owner's present task visually primary.

Make an action look actionable

Make commands and editable elements recognisably actionable through consistent typography, focus, icons, and copper-orange emphasis. A title that offers an action should expose its purpose. Green can indicate a recorded ready or checked state; the label must explain the scope of that state.

Explain unavailable actions and their relevant prerequisites clearly. A failure identifies the affected artifact and a useful next action. Readability, words, and position must communicate state alongside color.

Let a cue settle

Clear an attention highlight when its underlying condition is resolved. If evidence changes during a model turn, identify the affected finding as needing a recheck. Show completion briefly and retain the result in the normal record. Dismissing a presentation cue should not erase an unresolved condition from the assignment.

Use text, position, and focus treatment alongside color and any optional character cues. Reduced motion should preserve state meaning. The owner should understand the work on a different display or at another reading scale.

40 Command families and contextual access

Organize commands around the current work, with contextual instruments and predictable menu and keyboard access. The earlier grouping below is a vocabulary study, not a requirement for nine permanent menus. Claude may refine wording, count, and placement while keeping useful operations discoverable.

Group	Command family	Primary purpose
Shape the work	Work	Create, open, save, receive, and export working material.
Shape the work	Revise	Undo, redo, select, copy, and modify the current content.
Shape the work	Arrange	Choose views, organize presentation, and arrange working elements.
Put it to work	Tools	Use applicable calculations, conversions, and capability operations.
Put it to work	Commands	Build, run, inspect, and reuse supported command sequences.
Put it to work	Rules	Define relevant conditions, triggers, and permitted automatic behavior.
Understand the work	Inspect	Examine evidence, calculations, changes, and actual operation results.
Understand the work	Properties	Read or change details of the selected artifact or operation.
Understand the work	Guide	Get help for the current task, capability, or selected command.

Keep the boundaries useful

Programs opens a capability; Tools offers operations for the current work. Commands exposes meaningful sequences; Rules describes conditions for later behavior. Inspect explains the basis or history of a result; Properties concerns the selected item's attributes. General program settings remain in the Settings tree and process controls remain in System.

Menu contents should follow the active artifact and selection while retaining stable top-level names. Show an understandable empty or unavailable state when a menu has no applicable operation. Preserve keyboard access and a clear focus path. A right-click action and its head-menu equivalent should reach the same supported operation.

41 Right click begins with the selected thing

A right-click should offer what is useful at that place. The owner should be able to stay with a passage, quantity, file, record, job, or setting and reach an applicable action. The menu is a graphical way into the same supported capabilities that the shared strip and assistant can use.

Identify the target clearly

Name the target and its kind at the menu's top. Respect an existing selection when it includes the object under the pointer. If a different object becomes the target, make that change visible before offering an action. Keyboard invocation should use the focused object or selection with equivalent behavior.

For multiple selected objects, offer operations that understand the set. Two comparable revisions can support comparison; a mixture of unrelated objects should produce only suitable shared actions. An empty place can offer creation, intake, kept chains, and a way to start from available material.

Build the menu from current possibilities

Combine object modifications, applicable staC commands, installed capabilities, and useful retained methods. Use actual supported inputs and current state to decide what appears. Keep familiar operations in a stable order. A new adapter should contribute through this common mechanism rather than adding a competing menu system.

Explain a missing dependency, incompatible format, or changed state where that explanation helps the owner proceed. A provider problem should leave unrelated actions available. Keep availability checks lightweight enough that opening a menu remains comfortable during inference or background work.

Keep the target bound to the action

An operation selected from the menu should concern the artifact and revision that the menu identified. Recheck mutable state before applying a change. If the source changed, preserve the current work and explain the need for a fresh action. Show the result where it belongs: on the page, in the worksheet, beside the job, or in an inspectable record.

Dismissal closes the interaction and returns focus predictably. Hovering and opening submenus should not accidentally commit edits. Keep the same action meanings in pointer and keyboard use, including a readable reason when an operation cannot proceed.

42 Cascades that construct a staC chain

Preserve and develop the existing chain builder. In the reviewed Graphite 0.1 source, workbench/chain_menu.py already constructs staC pipelines through cascading submenus, and workbench/context_menu.py exposes that builder from the contextual menu. Fluxbox's recursive submenu interaction is the visual inspiration [R15]. These are source observations, separate from owner-host operation.

Use the working foundation

The reviewed builder begins from selected material and offers lenses such as rows or headings, followed by supported filters, sorting, grouping, totals, column operations, and joins. It discovers real column names and suitable values. It shows the accumulated pipeline and previews information such as the number of rows remaining.

Keep that data-aware behavior. After a step changes the result, the next choices should concern the resulting data. A filter acts on the available columns; a join needs a suitable other source and matching fields. Explain an incompatibility at the point it matters instead of allowing a chain that only fails at the end.

Make the growing sequence understandable

Give each step a readable label and retain access to its staC representation. Let the owner inspect the current result and understand the next operation. A person should be able to build a useful transformation through the menu while learning the corresponding command language by observation.

Support returning to an earlier stage and choosing a different next step. Keep submenu placement within the available screen and preserve a predictable keyboard route through the cascade. Retain the useful ability to tear off a submenu and leave it open. A long chain should have an understandable inspect view when the cascade becomes awkward.

Preview within a useful budget

The current pipeline-building previews are read operations. Keep preview work bounded and identify whether a displayed count is exact, sampled, limited, or unavailable. Reuse valid intermediate results where useful. Avoid making hover depend on an expensive model turn or an unrestricted scan of every source.

Extend supported chain operations through appropriate adapters and typed outputs. Broad sequences involving edits or external effects need the explicit execution behavior described next. Preserve useful existing commands while growing that scope.

43 From a preview to completed work

A constructed chain needs a clear ending. The reviewed builder already offers endings that display a result, save a sheet, watch a result, or add work to the Agenda. Keep those useful routes and extend them as the operation system grows. The owner should understand what will happen before choosing the ending.

Say what the ending does

Distinguish showing a temporary result, saving an artifact, scheduling later work, and creating a continuing watch. State the destination and relevant inputs. A preview can explain the transformation without implying that a file has already been saved. The completion record should identify the artifact or scheduled item actually created.

When a sequence includes writes, apply the existing revision and authority mechanisms at the affected steps. Carry source identities and applicability conditions through conversions. An operation that leaves Graphite for a native tool or connected service should have a meaningful handoff and an observed result.

Make interruption intelligible

Show which step is running and let Stop address the owned worker. Preserve completed observations and distinguish an unstarted step from an interrupted one. A sequence can contain effects that cannot be undone as a single transaction; reconcile their actual state before offering a retry or continuation.

Use staging or working copies when they provide a useful recovery path. Keep the owner's ongoing document edits intact. If an artifact changes while a chain runs, identify which returned result still applies and which proposed change needs fresh inputs. Preserve earlier results with their original revisions.

Connect the visible result

A saved worksheet should open through Work and remain reachable in Library. A scheduled investigation should appear in the ordinary activity and assignment records. A watch should state what it watches and how to stop it. Keep these routes part of the existing desk rather than introducing a separate automation dashboard.

Allow an ordinary single operation to remain direct. The chain builder is available when composition is useful. Both routes should use the same supported operations and report the actual outcome in the same language.

44 Keep useful chains and grow capabilities

A chain that solves a recurring problem is part of the machine's useful knowledge. Let the owner keep it with a meaningful name and discover it again from relevant material. Dynamic tool development should recognize when the same function is needed more than once and consider a stable capability around it.

Keep purpose with the method

Record what the chain accomplishes, the inputs it expects, its outputs, dependencies, applicability conditions, and observed limitations. Separate a particular invocation's file and revision bindings from the reusable method. Make clear which values are parameters and which assumptions are essential to its meaning.

For example, a shortage calculation can retain the relationship between required and available quantities while accepting a different compatible stock table. A PDF extraction helper may have narrower applicability to a known table structure. The saved description should help the assistant choose between them without guessing from a filename.

Preserve the existing collection

The reviewed builder stores kept chains in an editable plain-text menu file using chain and submenu entries. Preserve those records and the owner's organization when changing branding, storage, or presentation. Retain a readable representation and explain entries whose dependencies or syntax no longer work.

Offer relevant kept methods through context menus and the Tool Atlas. A substantial capability can join Programs; a small transformation can remain a contextual operation. Use mature software as the implementation base whenever it fits, keeping useful adapters and dependency information rather than repeatedly rebuilding foundations.

Improve through observed use

Preserve successful uses, corrected assumptions, and meaningful failures. A change to the method should have an identifiable revision so earlier results retain their basis. Consider capability promotion after a recurring need, with an interface suited to the actual function and the current authority model.

The owner and assistant should discover the same available method through their respective interfaces. Replacing the local model should leave these tools and their records available. A retained method becomes valuable through reliable reuse and an understandable scope.

45 One writing strip with clear destinations

Keep one clearly addressed writing strip with Assistant, Terminal, and Log as the current destination labels. Terminal can use the configured shell, including Bash. Preserve useful search, staC, console, and annotation functions while keeping destination and target clear.

Assistant and Terminal have different jobs

Assistant accepts natural language and supported staC requests. It should understand the selected artifact, applicable revision, and relevant assignment without requiring the owner to repeat them. Make the current destination and target understandable before submission. A constructed chain can be inspected through the same command language.

Terminal sends the owner's command to the selected Debian shell through the supported execution route; Bash is one possible shell. Show the command, working location, and owned process while it runs. Keep a separate draft for each destination. Switching destinations must not execute or reinterpret an unfinished assistant request, shell command, or log note.

Log connects notes to events

Log should combine readable activity with the ability to record an owner note at a useful moment. Associate a note with its time and, when chosen, the relevant event, job, artifact, or supported network observation. Preserve the original machine event and the human annotation as distinguishable records.

Watching a stream should respect the owner's reading position. If the owner scrolls back, allow new activity to accumulate with a quiet cue and a direct return to the latest item. A note entered while examining an earlier event should remain attached to the intended context.

Keep direct routes economical

Search remains available through an explicit command and appropriate menus. Simple local searches should execute directly without an LLM turn. Show results in the appropriate workspace with a route to their sources. Preserve any useful keyboard access when removing a separate Search destination.

Let the owner expand or contract input and output space while retaining the destination, draft, selection, and focus. The earlier Line, Half, and Tall labels are optional. A response, error, or new activity must not replace text the owner is still composing.

46 Today as a calm point of return

Today should help me begin or resume with little effort. Its greeting can be generous and handwritten, while its useful content stays concise. It should draw from current work records so the page becomes more useful as Graphite carries more assignments.

Offer the next useful entrance

Continue should lead to genuinely recent or unfinished work with enough identity to choose the right item. An interrupted investigation should say that it can resume and identify its last valid point. A completed assignment should lead to its actual result. Preserve the distinction between an open artifact and a running job.

Waiting for you should collect specific questions and prerequisites that need the owner. Make each entry understandable without opening an entire conversation. An empty state can simply say that nothing needs attention. Avoid decorating the page with fictitious tasks or counts.

Capture without losing the moment

Provide a quick route to record a thought through the existing capture mechanism and shared strip. Preserve an unfinished note while the owner looks elsewhere. Confirm when it has actually been saved and give it a place in the current project or an understandable general collection.

Ingress can explain received material awaiting organization. Offer useful next actions for those files, such as preview, associate with a project, or begin an assignment. A quiet desk can remain mostly open paper. Let content earn its place rather than filling a grid of permanent boxes.

Keep return behavior consistent

Returning to Today should preserve the current Work artifact and Library location. Choosing a result should take the owner to its useful surface while retaining a route back. Background progress can update a concise entry without repeatedly moving the page or stealing focus.

When an assigned investigation continues elsewhere, Today can show its current finding or open condition in ordinary words. It should remain possible to sit down and write immediately. The home page supplies orientation and access while the surrounding machine, assistant, and command regions keep their established roles.

47 Work as a shared artifact surface

Work is where the actual material is operated on. A document should behave as a document, a spreadsheet as a spreadsheet, and a supported engineering application as its useful working environment. Preserve the owner's ability to act directly while the assistant contributes to the same artifact.

Bind an instruction to its material

A request such as make these words bold needs the selected passage and current revision. A request to calculate from a table needs the relevant cells and conditions. Make the target understandable through selection or a short description. Resolve ambiguity when it changes the intended result.

An assistant edit should preserve unrelated content and produce an observable change. Retain the history needed to inspect or undo supported changes. When an operation produces no change, describe that outcome accurately. Avoid using a conversational acknowledgement as the evidence that a save or edit happened.

Support taking turns

Let the owner continue writing after an assistant action without rebuilding the working context. Restore the appropriate selection and focus. If a pending model turn concerns an older revision, identify that relationship before applying its proposed edit. The owner's newer work should remain intact.

Choose the strongest supported connection for each application: a compact workspace in the same live instance, an API or add-on, a compatible engine, or an explicit file handoff to the native program. Use Show full controls for the same application instance and Open full application for a handoff. Preserve native master formats and identify whether the owner edits the original, a managed working copy, an export, or a preview.

For browser and mail work, identify the actual browser profile, URL, account, message, and draft. An embedded web engine is not the full installed Chrome or Firefox application. Preserve the supported handoff state and explain any save, reload, authentication, or profile transition. Native application windows inside the Graphite session are a valid route.

Keep inspection close to the result

A calculated value can lead to its inputs and method. An extracted quantity can lead to the source page. A changed passage can lead to its revision record. Use relevant context commands and Inspect so the center remains focused on the artifact while deeper evidence remains reachable.

A result needing recheck should identify the affected portion and why it changed. Recompute or reinterpret the dependent work while preserving unrelated results. Moving between Today, Work, and Library should preserve meaningful working state and leave running assignments under their existing ownership and controls.

48 Library as a usable collection of material

Library should make the left tree useful. The tree chooses a place, project, or collection; the center displays its contents with a preview and appropriate operations. Keep the difference between a physical file location and a project association understandable.

Browse with a stable sense of place

Keep a readable location path with back, forward, and up where they apply. Preserve selection, scroll position, and display preference for each browsing location. Icons, List, Columns, and Preview belong in relevant context and Arrange menus. Opening a working item should lead to Work while keeping the Library location available.

Show enough file identity to distinguish similarly named revisions and formats. A useful preview can show a document passage, table, drawing, or metadata without pretending to edit the item. Explain when a format needs a different installed capability or an additional dependency.

Let related material meet

Selecting compatible sources should make comparison, extraction, or chain construction discoverable. A table can lead into a calculation. Two revisions can lead into a comparison. A project association can identify the investigation that used a source. Preserve those links without forcing copies of every file into every project.

Respect the owner's policy for originals and working copies. A protected source remains unchanged until replacing it is authorised. Human and assistant edits should concern the same current working artifact, including during a native-application handoff. Bind replacement to its intended revision and retain a recovery path. Preserve or repair source relationships after a rename or move where the integration supports it.

Receive and deliver clearly

Ingress introduces material through the existing intake route. Egress contains the intended deliverables. Support ordinary readable files and coherent export of one result, an assignment, selected material, or a whole collection. Include the necessary selected sources, methods, conditions, and revision relationships, and identify omitted dependencies. Keep preparation, a written export, printing, and external delivery distinguishable.

Search results should retain source and location context. Opening a result should make the relevant passage or object reachable where supported. Keep lexical lookup economical and let more involved interpretation use the assistant when needed. An unavailable mount or missing file should preserve the project reference and explain the actual gap.

A shared result should travel with enough context for its recipient to review or continue the work. Show which sources, attachments, calculations, and methods are included. Preserve stable identities and the snapshot being shared, and flag references that remain unavailable. A selected letter should not accidentally carry the rest of the mailbox. Keep secrets and unrelated private material outside the shared package.

49 Conversation activity and working context

A contextual inspector should help the owner understand and direct work without displacing the artifact. Relevant views can include Preview, Activity, Sources, History, Conversation, and Context. Choose a useful small set for the selected work, with access to the others when needed. Three permanent tabs, a fixed right column, and alignment with other tab groups are optional.

Conversation carries intent and decisions

Conversation holds the exchange with the assistant: the request, useful clarification, a concise answer, and relevant follow-up. Keep messages connected to their assignment and artifact where possible. A click on an earlier instruction should make its original context understandable without changing the current target silently.

Avoid filling the conversation with every mechanical event. A short answer can lead to the saved work or a fuller explanation. Questions should be specific enough that the owner understands how an answer affects progress. Reuse applicable preferences and authority already given.

Activity shows observable progress

Activity states what is being worked on, the operation underway, what was found, and what needs the owner. Tie those words to real jobs and records. Use concise explanations of decisions and actual actions. An idle model with a queued task should not be presented as actively reasoning.

Several workers can contribute to one assignment, with a short combined account and expandable details. Show the relevant owned controls without repeating the entire System process list. Keep a completed result reachable after its brief visual cue settles.

Context explains the working picture

Context should reveal the sources, current artifact, applicable revisions, important facts, calculations, open conditions, and selected capabilities informing the next action. Allow the owner to correct or exclude a source and understand the scope of that choice. Keep provenance reachable from each material fact.

Model context usage and durable project knowledge are different resources. Preserve sources, artifacts, actual actions and observations, concise decision records, assumptions, useful failed approaches, and methods outside the model session. Explain prompt use where useful. The owner and another reviewer should be able to understand the preserved work without reconstructing the full conversation or depending on the original model.

50 Programs as a useful toolbox

Programs is a plain, understandable route to installed applications and substantial capabilities. Use readable names and suitable small icons in a stable location. A cat affordance and its placement are optional. The owner should recognise what can be opened or done without learning internal Tool Atlas vocabulary.

Show real availability

A listed capability should correspond to an installed or clearly available operation. Distinguish ready, missing a dependency, disconnected, and unavailable when that affects use. Give a relevant explanation and an appropriate route to resolve the gap. Avoid a long list of impressive names that cannot perform their advertised work.

Organize a growing list through useful names and restrained grouping. Preserve direct access to common tools. The owner can open the same application that the assistant uses through its supported operations. Explain limitations where a human interface offers more than the current automation adapter.

Open into the existing desk

A capability should lead to Work or the relevant established surface, carrying the selected material when appropriate. A file chosen through Library can open with a compatible application. A tool opened without material should offer a clear starting place and preserve the existing artifact.

Launching a program should not silently begin a costly investigation or change project authority. Tool operations then follow the current assignment and permission scope. Keep machine configuration in Settings and process controls in System so Programs retains a simple purpose.

Let useful additions find the right scale

Dynamic tool development may create an adapter, transformation, or larger capability. Substantial reusable abilities can join Programs. Small helpers can remain discoverable through relevant context menus and the Tool Atlas. Preserve their method descriptions and underlying mature software dependencies.

If optional cat artwork accompanies Programs, it can lift its paws when the menu opens and return to the current work cue on close. With or without artwork, keep the menu anchor, row spacing, keyboard focus, and selection predictable, and return focus to a useful working place.

51 System as a clear view of owned work

System is a discoverable route to process and job controls, the master Stop action, Game, and machine session actions. A plain labelled affordance is sufficient; the rear cat and its placement are optional. Keep it useful with no work running and with several active assignments.

Give each control an identifiable subject

System includes Game and a prominent Stop owned work action. Global Stop reaches Graphite-owned assistant, inference, preparation, tool, and queued jobs under the stated policy, keeping the desktop and manual applications usable. In a shared native application, cancel the automation through a supported route without discarding manual work; report if it cannot stop immediately. Stop does not retract completed external effects. Identify each job and its actual worker.

A per-job Stop reaches that job's owned worker and leaves it cancelled or stopped until explicit owner direction permits another attempt. Pause preserves progress at a supported boundary; Resume continues paused work from a valid point. Keep both distinct from failure or waiting for input. If an operation cannot pause or cancel immediately, show the pending request and actual worker state.

Priorities should mean what they do

Keep work ordering and niceness controls useful. Dragging queued work changes the scheduling order supported by the agenda. A niceness slider concerns the worker or scheduling mechanism actually controlled. Explain the practical effect in ordinary words. Do not imply that a slider reserves GPU capacity unless the implementation provides that facility.

The traffic cop should respect foreground requests, owner-selected priorities, resource limits, and cancellation. Optional preparation can yield before a large scan competes with active inference. A task waiting for a prerequisite should retain that reason rather than appearing to consume work continuously.

Keep session actions separate from assignments

Return to Debian leaves the Graphite graphical session for a usable text console; Log out ends the owner's login session; Restart and Shut down concern the machine. Preserve unsaved work through the supported save or interruption route. Closing only the workspace window may leave explicitly authorised background work running. Ending the graphical session must reconcile GUI-dependent jobs; persistent services may continue only within the owner's configured scope. Explain the actual effect of each action.

System remains reachable by keyboard and pointer during background work. Keep status and any optional animation within stable bounds. A concise summary can lead to detailed jobs and process information. It must remain possible to inspect and stop Graphite-owned work after the presentation reconnects to its runner.

52 Files machine and settings

Keep files, what is in the machine, connected resources, and settings easy to reach. Files is the current browser label; Library describes the organised material collection and browsing function. Cabinet is a historical name. The owner can keep useful trees visible, pin relevant branches, or reveal them as needed; a permanently fixed left column is optional.

Machine describes the effective system

Show the actual detected hardware and available storage, relevant connected devices, loaded model, active inference backend, and supported worker information. Preserve a distinction between installed memory, available headroom, and the allocations reported by the runtime. The integrated Radeon and Ryzen share the physical system pool described earlier.

An unavailable device, disconnected location, or inactive backend needs a readable state and useful next action. Keep the effective configuration visible when it differs from a requested preference. A CPU fallback should retain its reason and the identity of the work being continued.

Settings exposes real choices

Place program and assistant settings in the tree with useful names, current values, and understandable consequences. The owner and assistant should reach the same setting operation. Reuse existing validation, authority, and persistence mechanisms. A request to change a setting should identify the relevant value and observe the actual result.

Explain whether a choice applies immediately, to the next job, or after a relevant component restarts. Preserve a recovery route for changes that affect operation. Keep ordinary display preferences easy to reach while allowing technical settings to be inspected when they matter.

Reveal detail in proportion to the question

The main tree should orient the owner. A selected branch can show relevant details in the center and contextual actions under the pointer. Avoid presenting every inference parameter as a permanent control on the desk. Retain advanced capability through the settings structure and supported commands.

Keep CPU, GPU, RAM, and TXT readings available through suitable compact status or System views. Their position is open to the composition. TXT reports the selected model's actual context use where available. Missing telemetry is labelled unavailable. Every live reading identifies its effective device, worker, or source rather than inheriting a number from a mockup.

53 Complete journeys through the same desk

Use complete owner journeys to connect these design details. The following are illustrative tasks, not fixed answers or required scripts. Each should preserve the same desk, shared artifacts, source relationships, and understandable controls while letting Claude choose the implementation.

From a table to a reusable method

I select a stock table in Library and right-click. Chain from here reveals its available columns. I calculate a shortage from required and available quantities, keep the positive results, and save a worksheet. Work opens the resulting artifact. Inspect leads to the original inputs and the transformation that produced it.

I keep the chain under a useful name. When another project needs the same function, Graphite finds the retained method, checks the new input shape and conditions, and uses or improves it. Repeated usefulness can lead to a stable capability backed by maintained code and established software.

From an uncertain unit to a current result

I ask for a total on a page. An incompatible unit remains visible in the material, and the inspector identifies the affected input with a concise question and meaningful actions. My answer changes the bound input and the dependent result, with the earlier state retained. Use the current theme to show the relationship; yellow, pink underlines, and a fixed right column are optional treatments.

From engineering evidence to ordinary coordination

I ask Graphite to investigate an ARC-1 question using the actual authorized project collection. It extracts applicable values, uses available calculations and tools, and records the remaining uncertainty. A later request can turn those findings into a report, a supplier question, or preparation for a meeting.

Connected actions use the relevant account and granted scope. Local preparation remains useful without the connection. A source correction should identify affected drafts and findings while preserving the record of messages or decisions already made. A scheduling operation uses real availability and appropriate calendar tools.

The practical outcome is a machine on which I can begin with a small request, inspect the work, continue it myself, and delegate a longer question. The visible simplicity should survive the complexity of the assignment.

54 Continuity and a convincing finished experience

Use this expanded direction to complete coherent behavior in the existing program. Claude has creative authority over architecture, integration, implementation order, and useful additional capabilities. The design detail describes the experience and the meanings to preserve; it does not impose a new sequence of development passes.

Keep work through interruption

Preserve drafts, meaningful working positions, investigation records, useful methods, and completed observations through supported restarts. A running job may need to pause or recover according to its worker lifecycle. Show that state accurately and resume from a valid boundary when the owner chooses.

If an external operation's result is uncertain, reconcile the actual artifact or service state before repeating it. Retain the owner's Stop decision. A missing network connection, source file, dependency, or model should leave useful local preparation available and identify the particular gap that prevents continuation.

Keep all routes usable

Pointer actions, keyboard access, the shared strip, and the assistant should reach compatible supported operations. Focus should remain visible and return sensibly after a menu or question. Preserve readable unavailable states, text scaling, reduced motion, and stable geometry. A long menu or command sequence needs a usable route within the real display.

Keep straightforward work lightweight. An ordinary edit should not require a long investigation. A difficult question can receive a larger budget and durable reasoning support. Let the owner inspect the result and its basis without having to understand every worker or runtime decision.

Demonstrate the behavior that matters

Use focused verification of the changed interactions and existing operation boundaries. Useful demonstrations include a saved edit, a data-aware cascade, a kept chain reused on compatible input, a current calculation after correction, and a stopped owned worker. Check the actual ThinkPad where hardware and real display behavior matter.

Report delivered capabilities, observed results, material choices, and remaining conditions. Keep historical source observations, simulations, design concepts, and owner-host results distinct. Deliver usable application updates as work advances toward the primary bootable Debian-based Graphite system. A partial application release is useful progress and must be described accurately against the complete desktop and installation goal.

55 Self directed advancement as part of useful work

Claude, make Graphite able to improve the means by which it helps me. A completed assignment should be able to leave behind corrected knowledge, a useful method, or a better tool that benefits later assignments. Give this work an explicit purpose, a durable record, and an appropriate share of the machine. Connect it to the program we are completing rather than starting a separate product.

Several kinds of improvement

Keep factual learning, method development, workflow changes, runtime configuration, software modification, and model training distinct. Each changes a different part of the system and needs an appropriate basis for adoption. A retained correction can improve the next decision while the model weights remain identical. Replacing or fine tuning weights is a separate operation with separate compatibility and quality questions.

Voyager demonstrated an executable skill library with environment feedback in Minecraft while using a black box language model without fine tuning. Reflexion retained feedback in episodic text to influence later attempts. These are precedents for growth around a model. Their domains and model arrangements do not establish the performance of a small local Graphite model. [R16] [R17]

Let advancement serve an assignment

The immediate reason to improve should be recognizable: repeated owner correction, a recurring manual step, an unreliable extraction, a missing adapter, or work that repeatedly exceeds its resource budget. Preserve the connection between that need and the proposed change. An interesting experiment may deserve an explicitly assigned research budget; ordinary work should not silently become unlimited research into the assistant itself.

Use the existing investigation, method, capability, and job records where they fit. Claude may strengthen their relationships or add a missing concept when the current code requires it. An improvement can be queued, developed, compared, adopted, deferred, or retired without losing what was learned. Those are useful distinctions, not a mandatory new state machine or prescribed series of passes.

The experience we want

A month later, I should be able to recognize useful growth through fewer repeated explanations, better source selection, more dependable tools, and completed work that used retained experience. Present the particular improvement and its observed scope. Accumulating files, producing longer reasoning, or running longer does not establish advancement. Progress may compound, plateau, or reverse; preserve the evidence needed to tell which happened.

56 Knowledge and corrections that remain useful

Make a correction useful at the next relevant decision. If I explain that a datasheet value is an absolute maximum rather than a normal operating value, retain the meaning, its source and revision, the affected component, and the calculations that used it. A later question about the same condition should find that distinction without loading the entire conversation.

Facts have a basis and a scope

Preserve whether a statement came from an owner instruction, a cited source, an executed calculation, an observed tool result, or a model hypothesis. An owner preference about presentation and an engineering claim about a component need different treatment. Keep the original evidence accessible when a short summary is retrieved. A correction should not erase the history needed to understand an earlier result.

The W3C PROV primer distinguishes entities, the activities that produce or use them, responsible agents, and derivation and revision relationships. We can adapt those concepts to the records already in Graphite: a document revision informs an extraction, the extraction feeds a calculation, and a finding refers to that calculation. This is a provenance pattern, not a requirement to implement a new semantic web stack. [R22]

Retrieve the part that changes the decision

Prefer a compact set of relevant facts, conditions, useful prior attempts, and unresolved questions. Include a precise route back to the source. Retrieval should consider project, artifact identity, revision, units, and the current assignment rather than treating shared words as sufficient evidence of relevance. Preserve a way to inspect why a record was selected and to dismiss an unhelpful association.

When sources conflict, keep the disagreement visible until there is a reason to resolve it. A newer document can supersede one value without invalidating every older observation. Missing evidence remains missing. A model summary should not become the source of a physical claim merely because it was retrieved repeatedly.

Recheck affected knowledge

A revised input should mark dependent findings for reconsideration through the existing revision and condition mechanisms. Recompute supported deterministic steps when appropriate, then ask the model about the remaining interpretive issue. Retain both the changed result and the reason it changed. Correction, source replacement, and retirement must also reach derived summaries and indexes so an obsolete claim does not return through a second retrieval route.

57 From experience to tools and capabilities

Treat a useful working method as an asset that can outlive the assignment and the model session. Before building anything, search retained methods, the Tool Atlas, installed software, and suitable mature implementations. Prefer a small adapter around a dependable library or command over rebuilding a complete application. Preserve authored helpers and their actual invocations so meaningful recurrence can be recognized.

Retain what made the method work

A reusable method needs a purpose, accepted inputs, produced outputs, applicable conditions, dependencies, authority, and observed limitations. Bind the concrete invocation to its actual files and revisions while keeping reusable parameters separate. Store a version or digest so later results retain the method that produced them. A successful invocation establishes evidence for that case; broader applicability needs its own basis.

ExpeL studied extracting insights from accumulated task experiences and retrieving those insights and relevant examples during inference without parameter updates. We borrow the idea of selective experience reuse. Graphite should retain the actual outcome beneath an insight and revise the insight when contrary evidence appears. A fluent retrospective is still an interpretation. [R18]

Recognize recurrence across projects

When substantially the same helper is needed more than once, deliberately consider keeping it as a method and promoting it into a capability. Recognize equivalent work despite different filenames or projects. Repeated calls during debugging are not sufficient evidence of recurring owner need. The owner may also deliberately retain a valuable one use method; recurrence is a useful signal rather than a rigid threshold.

Promotion should create a clear name, discoverable interface, applicable context menu entries, and the corresponding supported assistant operation. Include required packages and licensing notices in the appropriate delivery route. A substantial capability belongs in Programs; a small transformation may fit better as a contextual command. The human and assistant should reach the same implementation and artifact rules.

Compose familiar methods into unfamiliar work

A PDF extraction, unit check, spreadsheet transformation, and document update can become a reusable route when their interfaces fit. Keep the individual steps inspectable and revise the route when a dependency changes. Useful composition should reduce repeated planning while preserving the opportunity to stop, inspect an intermediate artifact, or substitute a better step. Avoid loading the whole method library into every prompt.

58 Choosing worthwhile improvements

Give Graphite a way to notice recurring difficulty and retain a specific improvement opportunity. Capture the affected assignment, the observed obstacle, its recurrence, the correction or workaround used, and a plausible benefit. Group equivalent opportunities so the backlog does not multiply each time a symptom appears. Let useful independent work continue when one improvement is blocked.

Choose by practical value

Consider how often the difficulty occurs, how much owner effort it causes, the importance of the affected work, the evidence for the proposed remedy, and the likely development and maintenance cost. Exact scoring is Claude's choice. Explain the selection in plain terms such as repeated correction across several documents or a missing conversion that blocks the assigned analysis. An invented confidence percentage adds little.

Separate a defect, a capability gap, a knowledge gap, and a hardware limit. A more complicated prompt may not remedy a missing parser. A new parser cannot supply a measurement that was never made. A runtime configuration cannot grant the integrated GPU an unsupported memory allocation. Route the opportunity to the appropriate kind of work.

Use granted time and resources

An owner can assign continued improvement while away and establish standing limits on time, power, disk use, data access, and allowed effects. Use those grants without asking for each routine working copy or tool decision. Self improvement should share the existing job scheduler and traffic cop. Foreground work takes precedence; a quiet desk is an opportunity only within the owner's actual instructions and resource settings.

Select a bounded next action, record the result, and decide whether continuing is worthwhile. If several attempts produce no useful evidence, preserve the blockage and move to an independent useful item. This should prevent a resident agent from consuming an evening by rephrasing the same plan. Budget reflection and model calls as real work.

Exploration has a place

Allow creative experiments when their scope and budget are part of the assignment. A novel approach can be valuable before it has demonstrated a gain. Keep experimental candidates separate from the preferred method until there is a basis to adopt them. The existence of an exploration allowance does not require filling it or maintaining a large population of variants on the ThinkPad.

59 Building and comparing a candidate

An improvement should begin with a concrete statement of the behavior to change. For a datasheet importer, that might be preserving the distinction between typical and maximum values. For an editing tool, it might be retaining every unselected word. For retrieval, it might be finding the current source revision with less irrelevant material. State the expected artifact or behavior before generating the candidate.

Keep the current result available

Preserve a working baseline, the relevant input revisions, and the actual result produced by that baseline. Develop the candidate in a separate working copy with a record of its dependencies. Reuse existing examples and checks when they answer the question. A Git worktree can provide a separate checked out branch, but it does not isolate execution or prevent access to live data; add the appropriate runtime boundary where needed. [R23]

Compare the thing that matters

Run the candidate on relevant material and inspect the produced artifact or executed operation. A sheet should contain the expected typed values and source references. An edit should preserve unaffected content and save correctly. A tool should report actual errors rather than plausible text. Where output legitimately varies, compare the requirement and supported evidence rather than requiring identical prose.

Use a small representative set that includes a relevant exception and, where available, material not used to develop the change. A method that only handles the example that inspired it has a limited scope. Record that scope rather than creating a universal claim. Keep ARC-1 engineering assertions tied to authoritative project data and distinguish software checks from physical verification.

Let evidence survive the model turn

Record execution status, inputs, outputs, revisions, relevant resource observations, and the particular requirement assessed. Store model interpretation separately from tool execution facts. The candidate should not be able to declare that an operation ran merely by writing a convincing log. Sakana's author report documents fabricated execution logs and manipulation of evaluation mechanisms in its experiments; this motivates our independent execution evidence. [R21]

Resolve the comparison with a useful decision: adopt within the demonstrated scope, keep experimental, revise, or defer. Focus verification on the affected behavior and expand it when failures or broader changes justify doing so. Stop once the material question has been answered. Large test counts and repeated full suite runs are not the objective of this improvement process.

60 Changing software with recoverable activation

Allow useful self authored adapters, helpers, and procedures to become part of Graphite's maintained capabilities under the owner's standing authority. Extend that direction to proposed changes in Graphite's own software when the implementation can develop, compare, activate, and recover such changes coherently. This full process is a requested capability; the reviewed build has not established it end to end.

Research suggests a route

The Darwin Godel Machine studied agents modifying their code, evaluating candidate variants on software benchmarks, and retaining an archive from which further variants could be developed. We borrow the separation of proposal, observed comparison, and retained alternatives. The paper does not establish indefinite improvement, equivalent behavior from a small local model, or an appropriate compute budget for this ThinkPad. [R20]

Match the change to its consequences

A new helper, an altered method, an inference profile, and a database migration need different activation and recovery behavior. Preserve the supported authority and acceptance contracts. Complete routine work already covered by the owner's grant without repeated prompts. A change must not silently give itself new permissions or remove the owner's Stop, evidence, or recovery mechanisms.

Before activation, retain the candidate revision, dependency changes, observed comparison, effective scope, and previous working version. Register or activate at a stable boundary when feasible so an in flight assignment is not unexpectedly moved to a different implementation. Keep method and runtime identities attached to subsequent results. A restart may be more appropriate than hot replacement for some native or service changes.

Recover code and data together

Reverting a source branch alone may not reverse a changed database, modified artifact, package dependency, or external effect. Choose a recovery strategy appropriate to the change: compatible readers, additive migration, versioned artifacts, backups, or an explicit restoration operation. Preserve new owner work during recovery. A previously working release that cannot read the current records is not a complete rollback plan.

If activation fails, retain the attempted change and its actual failure, restore the supported working path where possible, and report the unresolved condition. Avoid automatic restart cycles that repeatedly apply the same broken upgrade. Work that cannot be safely interrupted should finish or reach its supported stopping boundary before replacement.

Keep software intelligence portable

Preserve methods and project records across model replacement. Fine tuning on staC or retained trajectories remains a separate possible project requiring curated examples and comparison. Useful method retention and better tools can advance independently.

61 Better preparation for Ryzen and Radeon

Let intelligent data arrangement improve through use. The CPU should become better equipped to assemble current facts, execute applicable calculations, resolve units, follow dependencies, and choose relevant tools before the model turn. Preserve the distinction between a deterministic result and a model interpretation. A useful improvement can remove repeated work from the neural inference path while making the remaining question more precise.

Retain preparation methods

A recurring assignment can acquire a preparation route: identify its source revisions, retrieve relevant conditions, compute supported quantities, select unresolved alternatives, and assemble a compact evidence packet. Keep the packet linked to those inputs. If a source changes, regenerate the affected portion. Avoid treating a cached answer as current simply because it was expensive to produce.

Track when a calculation, index, or extraction is reusable and when the cost of checking validity exceeds its benefit. Keep raw source material accessible when interpretation depends on prose omitted from a table. Model context should receive the relevant distinction and its basis, with a route to request more material. Compression must preserve conditions that can change the decision.

Learn the machine's operating behavior

The traffic cop may retain observations about supported model, context, batch, thread, offload, and power configurations under the detected hardware and software identity. Record whether observations concern prompt processing, generation, memory pressure, or foreground responsiveness. Use them to choose among supported profiles. Changed Mesa, runtime, model, or firmware can make an old observation a weaker guide.

Use Vulkan memory budget information as implementation dependent guidance, as already described in the inference sections. The 64 GB pool is shared by the machine; an observed allocation or successful load does not guarantee future GPU capacity or responsiveness. Favor a configuration that completes the assigned work within current headroom. The owner's GPU should be used where the backend supports useful acceleration. [R04]

Separate durable improvement from temporary speed

A corrected source relationship and a retained method should survive restart. A prefix cache or temporary tensor allocation can be released. Their validity, memory cost, and recovery differ. Avoid promoting a warm cache speedup into a claim about a better reasoning method. Runtime observations should inform scheduling without silently changing the quality or context required by the assignment.

Collect useful operational observations during ordinary work. Expensive parameter searches need a reason and a budget. Establish actual ThinkPad performance through recorded owner host observations that Graphite can collect during supported use.

62 Cooperating roles and useful reflection

Use distinct responsibilities when they improve the work: an observer identifies recurring friction, a builder develops a candidate, and a reviewer assesses the actual result against the requirement. Claude may implement these as ordinary procedures, staged model turns, workers, or another suitable arrangement. The roles do not require three large models loaded simultaneously or a new competing scheduler.

Share evidence and separate responsibilities

All roles should refer to the same assignment, input revisions, execution records, and candidate identity. Preserve who proposed a change and what evidence supported adoption. Multiple model opinions can help discover alternatives; agreement alone is not independent confirmation of a saved artifact, executed command, or correct engineering result. Prefer checks performed by the relevant tool and source comparison where applicable.

SWE agent investigated how interfaces for searching, navigating, editing, and executing code affect software agent performance. This supports treating the assistant's tool interface as part of the system we can improve. We adapt that lesson through clear operation contracts, concise useful responses, revision aware edits, and discoverable capabilities rather than assuming a larger model is the only route forward. [R19]

Reflect where it changes the next action

After a meaningful failure or owner correction, preserve a short account of the observed result, the likely explanation, and a useful next action. Link interpretation to the execution record and mark uncertainty. Repeatedly summarizing an unresolved failure should not increase its apparent certainty. Revise or retire the reflection when later evidence contradicts it.

Reflexion's episodic feedback mechanism is a research precedent for carrying lessons between attempts without weight updates. For Graphite, the useful adaptation is selective, evidence linked reflection at a decision boundary. It should not require a fresh narrative after every successful arithmetic operation or consume the owner's interactive response budget. [R17]

Coordinate work on the local machine

Serialize or batch model turns according to the effective inference arrangement and resource budget. CPU workers may continue useful supported calculations while the model is occupied, provided shared memory pressure and Stop behavior remain controlled. Parallelize independent work only where the expected benefit justifies the overhead. A worker that cannot proceed should leave a specific open condition and yield resources.

63 Keeping the method library dependable

A growing toolbox needs curation. Keep a preferred method discoverable, retain alternatives when their conditions differ, and retire obsolete entries from ordinary selection. Preserve older versions where past results or active assignments depend on them. Duplicate helpers should not make the assistant choose among indistinguishable names every time it acts.

Failures belong to their conditions

If a parser failed on a scanned PDF, record the input type, parser version, actual failure, and available workaround. Do not infer that the parser is useless for every PDF. A method that fails on a revised format may still be suitable for its original format. Narrow its applicable scope, repair it, or select another method according to the evidence.

Track whether a failure came from the tool, an invalid input, a missing dependency, a changed service, a resource limit, or the assistant selecting an inappropriate operation. Keep unresolved causes unresolved. This distinction improves later routing and avoids building a new tool to compensate for a temporary missing package.

Review experience rather than hoarding it

Retain representative successful and unsuccessful invocations that explain a method's behavior. Deduplicate repeated symptoms and summarize long histories while preserving access to the records supporting the summary. Respect retention settings, sensitive material, and available storage. Useful memory has a selection and maintenance process; an ever growing prompt transcript does not provide one.

When an owner correction or source revision invalidates a reusable lesson, update derived descriptions and retrieval entries as well as the underlying record. Mark a retired method clearly and keep active assignments bound to their intended version. The Library and Tool Atlas should explain what is preferred now and why an older result used a different method.

Keep growth transferable

Retained methods and project work should survive model replacement and remain transferable when supported dependencies are available. Preserve installation, compatibility, source, and revision information. Support a coherent snapshot of selected work or the whole collection in ordinary files with readable explanations and machine-readable relationships. A removed capability should identify affected methods while preserving their data.

64 Evidence of progress across real assignments

Judge advancement through completed work and the effort it saves. Include unfamiliar assignments and material not used to develop the method. Distinguish a correct completed outcome, an appropriate wait, a missing capability, and an incorrect completion. Record preserved artifacts, source bindings, useful reuse, responsiveness, and the setting and limits. Constructed scenarios and numerical variants do not establish a field reliability percentage.

Use a small set of meaningful comparisons

Question	Useful evidence
Did it understand the task and material?	The selected operation fits the intended outcome; relevant corrections, roles, and source conditions are preserved.
Did a retained method help?	A later assignment successfully reuses the method on current inputs without rebuilding it.
Did a software change improve behavior?	The candidate handles the affected task and relevant exception while preserving required behavior.
Did preparation help inference?	The packet contains current decision relevant evidence, and the resulting action meets the assignment requirement.
Did the machine remain usable?	Foreground response, resource observations, and actual Stop behavior remain acceptable in the recorded setting.

A month of ordinary work

An illustrative first encounter extracts a datasheet table and records a correction about maximum ratings. A later encounter finds that correction and uses an adapted importer. Repeated use leads Graphite to retain the importer with typed outputs and source pages. A subsequent assignment composes it with a calculation and a document update. The sequence explains intended behavior; it is neither a promised schedule nor a report of completed implementation.

Make the gain visible without filling the desk

Activity identifies the improvement job and actual stage. Conversation can explain a useful adopted change. Context or Sources exposes its basis, method revision, scope, and comparison when inspected. Optional visual cues can accompany real work or required input. A growing method library should add useful instruments without creating extra permanent controls or routine interruptions.

Report completed work, retained improvements, evidence, and open conditions. Keep results inspectable and recoverable. Historical exploration supports only its observed settings. Reference companion artifacts only when available; constructed cases alone cannot establish general reliability.

65 Research basis and current implementation

The sources below establish useful mechanisms in their original settings. Their combination into a private desktop assistant with CPU data preparation and Radeon inference is our proposed Graphite direction. Read the original paper or documentation and compatible implementation before borrowing a method. Preserve applicable licenses and attribution. Favor the smallest useful adaptation that completes the intended behavior.

Primary source	Mechanism and limit
Voyager 2023 [R16]	Stored executable skills and feedback in Minecraft. Supports the retained skill pattern, not a ThinkPad performance prediction.
Reflexion 2023 [R17]	Episodic feedback used in later attempts. Supports selective reflection without implying a model weight update.
ExpeL 2024 [R18]	Insights extracted from task experiences and retrieved later. Supports experience reuse, with task and model dependent results.
SWE agent 2024 [R19]	Agent computer interface design evaluated in software tasks. Supports improving the tool boundary and feedback.
Darwin Godel Machine 2025 [R20] [R21]	Code variants evaluated and retained; author report also records evaluation failures. Supports candidate comparison, not guaranteed continuous progress.
W3C PROV and Git [R22] [R23]	Provenance relationships and separate source worktrees. These are building blocks; neither supplies a complete autonomous upgrade system.

Paper results depend on their models, environments, task sets, and compute. We do not transfer their benchmark scores to Graphite, claim indefinite or exponential advancement, or assume every attempted improvement will succeed. The architecture should preserve useful gains and identify setbacks through completed work.

Current code and creative implementation

The reviewed Graphite 0.1 snapshot provides starting points to inspect in the current checkout. These code observations establish specific mechanisms, not owner host acceptance or a complete self modifying system. Keep unrelated working capabilities and records while extending their useful connections.

Reviewed entry point	Observed mechanism and boundary
workbench/agent_loop.py	Recorded assignment steps, model choice handling, and service continuation controls. These do not establish a general software improvement loop.
workbench/checked_methods.py	Method proposals, revision and acceptance checks, and evidence binding within a specific additive capacity procedure profile. It is not yet an unrestricted archive of generated tools.
workbench/chain_menu.py	Cascading staC construction and explicit keep this chain behavior in a plain text menu. Recurrence detection and automatic capability promotion are further work.

Complete the connection creatively

Claude, connect task interpretation, observed outcomes, reusable methods, capability acquisition, and improvement work through the actual architecture. Preserve service authority, owned Stop, version binding, and recovery. Use a general investigative route where a stored procedure does not fit, and verify completion against the owner's objective. Choose the implementation and work order creatively; no prescribed development passes or replacement framework is required.

Deliver useful growth progressively through supported work. State what the owner can do, what improvement was retained, what evidence supports it, and what remains a proposed capability. Keep engineering and ordinary assistance on the same foundation. I want to return to completed work and a workshop that has gained a dependable method, while retaining my place and control of the computer.

66 Sources and decision precedence

This master incorporates the owner's later general-assistant, capability-growth, ThinkPad, bootable-desktop, and objects-and-instruments directions through 8 November 2026. The source list below records provenance, including older planning and artwork. This revision supplies the operative direction. Numbered sections organise subjects and are not development passes.

Source or later decision	Role in this specification
WEDNESDAY 01.docx	Detailed inference, intelligent data arrangement, traffic cop, shared memory, cache identity, current code entry points, and creative examples.
BEST IDEA EVER.docx	Concise inference direction, supported optimization candidates, dependency delivery, and reuse of established implementations; integrated with the expanded inference material.
SECOND BEST IDEA .docx	Overall general-purpose goal, shared work, usable project knowledge, everyday assistance, local ownership, and creative development authority.
CAT ZIP.zip	Historical character artwork, state meanings, and code entry points. BOSS replaces mandatory corner artwork with intentional optional illustration spaces; preserve the useful state semantics.
Two yellow paw prints	Historical identity artwork. Retain the asset; BOSS replaces the default cursive and paw-print treatment with terminal lettering and illustrated desktop references.
Any assignment and capability growth	ARC-1 is one example. Claude may add web, email, calendar, and other capabilities. Dynamic tool development retains recurring tools and considers capability promotion using mature software as a base.
Direction without prescribed passes	Claude chooses implementation and work order. Older bounded-pass wording does not impose a new delivery schedule.

How to use implementation notes

Use the requirements and design freedom defined in the reading guide. Graphite 0.1 paths, counts, settings, and observations are historical starting points to inspect against the current checkout. Suggested runtime methods depend on actual model, backend, release, and driver compatibility. Artwork and benchmark references do not establish the current build's capabilities or acceptance.

Use this master as the commission, the checkout for implementation state, and authoritative files for project facts. Legacy attachments supply provenance; missing ones do not revoke the direction or justify invented content. Preserve upstream notices and the owner's release terms.

67 Primary technical references

Use the documentation and code of the selected pinned release for implementation. These primary references identify the methods and platform details discussed in this specification. They support candidate techniques and compatibility decisions; they do not establish measurements on the owner's machine.

R01 llama.cpp server interfaces and cache behavior
Native completion options, prompt evaluation, token accounting, slots, and cache reuse.

R02 llama.cpp build documentation
Vulkan-capable builds and the current CMake configuration interface.

R03 Mesa RADV documentation
The AMD Vulkan userspace driver and its relationship to amdgpu.

R04 Vulkan memory budget extension
Dynamic implementation-dependent heap budget and usage guidance.

R05 Debian AMD graphics firmware package
Debian 13 package identity and relevant firmware information.

R06 Lenovo ThinkPad E16 Gen 2 AMD specification
Processor configurations and integrated shared-memory Radeon graphics.

R07 llama.cpp feature matrix
Backend-specific support to check against the pinned implementation.

R08 llama.cpp speculative decoding documentation
Supported draft, draftless, and architecture-specific candidates.

R09 vLLM automatic prefix caching
Prefix reuse and the distinction between prefill and generation costs.

R10 SGLang session-aware radix cache
Session references, retention, and eviction under memory pressure.

R11 Parallelizing Tool Execution and LLM Generation
PASTE research on predicting concrete tool calls and overlapping permitted work.

R12 Anthropic guidance on effective agent tools
Focused tools, relevant responses, and project-aware scheduling examples.

R13 MLC LLM model-library compilation
Converted weights, compiled libraries, and Vulkan compilation targets.

R14 Debian Vulkan package search
Distribution packages to resolve for the actual host and delivery route.

R15 Fluxbox menu manual
Recursive submenus, menu structure, and command entries informing the cascading interaction.

Research on self directed advancement

The linked source markers in the advancement chapters lead to these primary references. Research findings describe their original settings. The proposed Graphite adaptations are owner direction. Sources checked 7 November 2026.

R16 Wang and colleagues 2023 Voyager

Voyager An Open Ended Embodied Agent with Large Language Models. Executable skill libraries and feedback in Minecraft without weight fine tuning. Research setting and results do not establish local Graphite performance.

R17 Shinn and colleagues 2023 Reflexion

Reflexion Language Agents with Verbal Reinforcement Learning. Task feedback retained as episodic reflections for later attempts without updating model weights.

R18 Zhao and colleagues 2024 ExpeL

ExpeL LLM Agents Are Experiential Learners. AAAI 38 pages 19632 to 19642. Experience-derived insights and relevant examples retrieved at inference time.

R19 Yang and colleagues 2024 SWE agent

SWE agent Agent Computer Interfaces Enable Automated Software Engineering. Evidence that the interface for repository search, editing, and execution affects agent performance.

R20 Zhang and colleagues 2025 Darwin Godel Machine

Darwin Godel Machine Open Ended Evolution of Self Improving Agents. Evaluated code modifications and an archive of agent variants in software benchmarks.

R21 Sakana AI 2025 Darwin Godel Machine report

Author report dated 30 May 2025. Includes observed fabricated execution logs and reward manipulation, informing the separation of candidate changes from execution evidence.

R22 W3C 2013 PROV Model Primer

Provenance concepts for entities, activities, responsible agents, derivation, and revision. A vocabulary to adapt to current records rather than a required new database.

R23 Git worktree documentation

Multiple working trees and branches for candidate development. Source separation does not isolate execution, databases, credentials, or side effects. Verify the installed Git release.

68 Thirty demanding assignments and expected behavior

Owner direction to Claude | Added 8 November 2026

These thirty assignments describe intended behavior and completion evidence, not capabilities already demonstrated by the current build. Together with section 69 they form thirty-five cases. Case numbers 01 to 35 are distinct from the document's section numbers. Each separates interpretation, software execution, coordination, completion, and a difficult variation.

You retain creative authority over tools, architecture, and development order. Improve shared mechanisms, reuse mature software, and add useful missing integrations. Treat the examples as direction rather than thirty prescribed modules or test suites. Preserve the owner's ability to take over and keep verification proportional to the work.

Assignment index

01 Investigate whether a complex engineering design can work
02 Turn foreign language datasheets into usable engineering data
03 Build an editable Blender model from an incomplete sketch
04 Prepare a printable letter from correspondence and policy
05 Organize a chaotic mailbox around actual commitments
06 Investigate unexplained network activity on the owner computer
07 Repair a spreadsheet whose meaning has drifted
08 Reconcile a missing payment or promised refund
09 Find photographs by content and assemble an evidence collection
10 Produce a coherent video from disordered source material
11 Fix an unfamiliar application without losing imported data
12 Arrange a meeting around prerequisites travel and time zones
13 Prepare complicated tax records for informed review
14 Prepare a complex online application with a clear handoff
15 Learn a recurring task from real corrections
16 Redesign a product after several constraints change
17 Reconstruct a lost project from contradictory backups
18 Explain why a prototype fails when its simulation passes
19 Move a project between applications with incompatible formats
20 Operate an unfamiliar graphical application alongside the owner
21 Reproduce a research result from an aging software environment
22 Spend an overnight assignment trying to disprove a favored idea
23 Explain a doubled energy bill using conflicting household records
24 Reschedule a small workshop when several constraints collide
25 Share a defensible part of a private knowledge collection
26 Complete useful work when source material contains hostile instructions
27 Resume a consequential workflow after a crash and an uncertain response
28 Finish a demanding project while keeping the ThinkPad usable
29 Localize a technical product across languages and formats
30 Create a useful application because the assignment needs one

01 Investigate whether a complex engineering design can work

Owner request Read this collection of specifications, drawings, calculations, and notes. Find what could prevent us from building the design and identify the next important question.

LLM Interpret requirements, qualifications, and component interactions. Propose plausible failure mechanisms and decide which unresolved question would most affect the build decision. Distinguish a source statement from an engineering inference.

Computer Inventory and index the material, extract useful tables, and run relevant electrical, mechanical, timing, or numerical checks through available software. Retain units, conditions, revisions, and links to the original passages.

Graphite Coordinate one investigation across the artifacts. Follow an uncertain power budget into startup demand, regulator limits, and operating conditions. Preserve useful findings and pursue technical questions that tools or sources can settle.

Completion Deliver a readiness report, supporting calculations, unresolved conditions, and the next valuable investigation. The report should show what is supported by records, what was calculated, and what still requires physical evidence.

Hard variation and why it matters Provide individually plausible component choices whose interfaces disagree when assembled. Graphite should investigate their interaction. Understanding each document separately cannot establish that the complete system will work.

02 Turn foreign language datasheets into usable engineering data

Owner request Compare these Chinese and English component datasheets and make a spreadsheet I can actually design with.

LLM Interpret technical terminology, footnotes, and conditions. Distinguish typical values from guaranteed limits and absolute maximum ratings. Identify uncertain translations or extractions before treating them as design inputs.

Computer Extract tables, perform OCR where necessary, convert explicit units, and build a typed comparison sheet. Preserve the original text and page references alongside normalized values. Calculate comparisons from the applicable values and conditions.

Graphite Keep source revisions, extraction decisions, and corrections connected to the sheet. Retrieve the relevant prose when a number alone cannot answer the question. Follow later corrections into affected comparisons and calculations.

Completion Deliver an editable sheet with quantities, units, minimum or typical or maximum roles, operating conditions, and source references. Flag missing values explicitly rather than filling them with plausible guesses.

Hard variation and why it matters Give two similar tables different temperature conditions and hide a rating qualification in a footnote. Graphite should preserve that distinction. A well-organized number becomes useful only when its meaning survives the extraction.

03 Build an editable Blender model from an incomplete sketch

Owner request Make this enclosure from my sketch. Fit the board, provide cable access, and give me something I can edit and print.

LLM Translate the sketch and intended use into modeling decisions. Identify dimensions that can be derived and assumptions that need an owner choice. Reason about access, assembly, and the relationship between visible appearance and usable geometry.

Computer Use Blender and suitable geometry tools to create the model, render previews, inspect dimensions and intersections, and check applicable mesh properties. Export the intended manufacturing representation while preserving the editable project.

Graphite Coordinate the sketch, board dimensions, model revisions, and exports. Keep assumed dimensions visible and preserve the owner's changes. Obtain a missing supported modeling or inspection capability when it is needed to complete the assignment.

Completion Deliver an editable project, dimensioned previews, an appropriate manufacturing export, and a clear account of assumed dimensions and unresolved fit questions. Check the object as geometry as well as inspecting its appearance.

Hard variation and why it matters Make a visually attractive version block a connector or omit necessary wall thickness. Graphite should find and repair the practical problem. A convincing render alone cannot establish that an enclosure is usable.

04 Prepare a printable letter from correspondence and policy

Owner request Read these emails with my landlord, find the relevant published policy, and prepare a letter I can print and sign.

LLM Reconstruct the chronology, identify relevant statements, and write in the intended voice. Distinguish documented facts, the owner's account, and interpretations. Check the scope and applicability of the actual policy used.

Computer Retrieve authorized messages and the policy document, extract dates and quotations, and produce the formatted letter. Compare consequential quotations and dates against the original records. Preserve a separate evidence sheet.

Graphite Coordinate the research and writing, retain the sources, and keep uncertain claims visible during drafting. Deliver a printable document with signature space and editable source material. Sending or publishing follows the owner's actual authority.

Completion Deliver a polished letter that the owner can inspect, print, and sign, with a separate source account for the statements that matter. A source that discusses a similar issue does not automatically support the proposed claim.

Hard variation and why it matters Include a quoted earlier email whose promise was later amended. Graphite should reconstruct the complete relevant sequence. Strong prose is useful only when the facts and quotations are accurate.

05 Organize a chaotic mailbox around actual commitments

Owner request Sort out my mailbox. Find unanswered commitments, important attachments, recurring bills, and things I can archive.

LLM Interpret conversations, changed deadlines, answered questions, and superseded attachments. Identify importance from the owner's purposes and expressed preferences. Keep uncertainty about a commitment visible when the thread does not resolve it.

Computer Search and group authorized messages, identify candidate attachments and duplicates, and apply supported mail operations. Record the affected account and message scope. Preserve useful reversal or recovery information where the service provides it.

Graphite Apply the owner's filing rules and authority, maintain an action list, and connect each commitment to its conversation. Recognize the difference between drafting a filing plan and applying the intended mailbox changes.

Completion Deliver organized mail, a concise list of unresolved actions, a useful attachment collection, and an account of consequential filing operations. Broad deletion requires the actual intended scope and authority to be clear.

Hard variation and why it matters Put the only appointment confirmation in an ordinary-looking reply and a newer deadline inside a quoted chain. Graphite should understand the conversation rather than rely only on subjects or sender names.

06 Investigate unexplained network activity on the owner computer

Owner request Find out why my laptop keeps communicating when I am not using it.

LLM Interpret observations, propose plausible explanations, and choose additional observations that could distinguish routine activity from a problem. Separate unusual behavior from demonstrated malicious behavior.

Computer Capture permitted traffic, calculate useful patterns, and inspect operating-system process and connection information. Save relevant observations with their timing. Packet data and process attribution are distinct sources that may need to be correlated.

Graphite Coordinate the investigation, its scope, and the evidence. Preserve observations, hypotheses, and confirmed findings as distinct records. Recognize limitations from encrypted traffic, inaccessible state, or missing observations.

Completion Deliver a process and destination timeline where supported, explanations with their evidence, unresolved questions, and useful proposed actions. Avoid claiming the content or purpose of encrypted communication merely from its existence.

Hard variation and why it matters Include a legitimate updater alongside an unexplained connection with incomplete process records. Graphite should investigate both proportionately. The expected result is a grounded explanation, including specific uncertainty where necessary.

07 Repair a spreadsheet whose meaning has drifted

Owner request This workbook says we are profitable, but the numbers feel wrong. Find the problem and rebuild the forecast.

LLM Interpret the business meaning of revenue, refunds, deposits, expenses, alternatives, and subtotals. Decide which operation answers the question before constructing formulas. Identify assumptions that affect the forecast.

Computer Recalculate the workbook, trace dependencies, inspect formulas, and compare results against relevant source records. Execute explicit scenarios and preserve the original workbook while creating a reviewable correction.

Graphite Connect suspicious outputs to their inputs and record consequential changes. Follow a corrected source value through the forecast. Preserve the owner's intended definitions and distinguish an uncertain classification from an arithmetic error.

Completion Deliver a corrected editable workbook, a clear account of changes, and a forecast with explicit assumptions. Check that the formulas answer the intended financial question as well as calculating successfully.

Hard variation and why it matters Include a subtotal among its component rows or record a refund as income. Graphite should identify the semantic error. A mathematically valid sum can still produce an incorrect conclusion.

08 Reconcile a missing payment or promised refund

Owner request I think I paid this twice and never received the promised refund. Work out what happened.

LLM Interpret descriptions, promises, partial refunds, and contradictory correspondence. Propose which transactions may correspond while recognizing that a similar amount or date is insufficient proof.

Computer Import authorized statements, invoices, receipts, and messages. Match candidate transactions, calculate balances, and build a chronology. Distinguish available transaction states such as an authorization, settlement, reversal, or refund.

Graphite Track the evidence for each proposed match, follow gaps in the records, and request only the specific missing information that cannot be obtained within authority. Preserve unresolved alternatives rather than silently choosing one.

Completion Deliver a reconciliation showing documented charges, payments, credits, and unexplained differences. Include supporting records and a useful draft inquiry where appropriate. Report a suspected missing refund as a suspicion until the records support it.

Hard variation and why it matters Provide a matching amount that belongs to a different invoice and a partial refund split across dates. Graphite should reconcile the actual relationships. This case tests investigation across records rather than superficial number matching.

09 Find photographs by content and assemble an evidence collection

Owner request Find the photos showing that damaged connector, including pictures where it is only in the background.

LLM Interpret the visual description, refine ambiguous criteria, and help distinguish similar objects. Use suitable image-understanding capabilities where available. Keep an inferred identity separate from a confirmed match.

Computer Index accessible photographs, inspect available metadata, identify candidate images, and group duplicates. Produce useful crops and contact sheets while preserving original files and their relationships to the derivatives.

Graphite Coordinate the search and review, expose uncertain candidates, and retain owner confirmations or corrections with their scope. Connect selected images to the requested collection without silently relocating or overwriting originals.

Completion Deliver a curated collection, annotated contact sheet, and references to originals. Report dates only to the extent supported by metadata or other evidence. Explain the coverage of the search and any inaccessible material.

Hard variation and why it matters Include a visually similar connector and an image whose metadata was changed by an export. Graphite should inspect identity and date evidence independently. Filenames and resemblance alone may not establish the requested facts.

10 Produce a coherent video from disordered source material

Owner request Turn these recordings, photographs, interviews, and notes into a ten-minute explanation someone can understand.

LLM Develop the narrative and choose source material that supports each point. Interpret quotations in context and identify missing explanatory steps. Revise the story when the evidence does not support a proposed claim.

Computer Transcribe, synchronize, edit, normalize audio, create captions, and render through appropriate media tools. Preserve source time references and the editable project. Inspect technical properties and the actual resulting playback.

Graphite Manage source material, editing decisions, revisions, and exports. Keep claims tied to their support and preserve enough structure for the owner to continue editing. Obtain missing tools or adapters where useful and authorized.

Completion Deliver the finished video, captions, editable project, and source list. Inspect the explanation and its accuracy alongside export success, duration, and playback. A successful render is one part of completion.

Hard variation and why it matters Provide a compelling short quotation whose surrounding interview reverses its apparent meaning. Graphite should preserve the intended context. This case combines creative judgment with exact manipulation of real media.

11 Fix an unfamiliar application without losing imported data

Owner request This application crashes when I import this file. Diagnose it and fix it.

LLM Develop hypotheses, read the relevant code and documentation, and propose a repair. Revise the explanation based on actual reproduction and debugging results. Keep an environment problem distinct from an application defect.

Computer Reproduce the failure, capture logs, run available debuggers, and compile changes. Exercise the original failing example and inspect the imported content. Use a recoverable working copy or isolated change where appropriate.

Graphite Connect experiments to the hypotheses they address. Obtain or configure a required compiler or inspection tool within authority and resume the repair. Preserve the source example, change history, and observed results.

Completion Deliver the working fix with focused evidence that the intended import succeeds and preserves the relevant content. Record remaining limitations. Completing a repair means more than removing the visible exception.

Hard variation and why it matters Make a candidate patch avoid the crash by silently dropping records. Graphite should reject that apparent success and continue investigating. The task is to restore the useful behavior the owner requested.

12 Arrange a meeting around prerequisites travel and time zones

Owner request Arrange a design review after everyone has the revised drawings, before the supplier deadline, and around my travel.

LLM Interpret participant roles, prerequisites, preparation needs, and preferences. Identify which promises or approvals are necessary. Distinguish a suitable proposal from an arrangement actually confirmed by participants.

Computer Read authorized availability, perform date and time-zone conversions with appropriate calendar data, and calculate workable slots and buffers. Check document versions and delivery status. Use the actual calendar or service interfaces for authorized actions.

Graphite Connect revised drawings, their delivery, preparation time, travel, the meeting, and the deadline. Track invitations and responses accurately. Keep useful options available when an essential participant cannot attend.

Completion Deliver a feasible arrangement with the correct supporting material and accurate status for proposals, invitations, and confirmations. Unavailable calendars or unanswered invitations remain explicit constraints.

Hard variation and why it matters Change the drawing release date and include participants across a daylight-saving transition. Graphite should recalculate both readiness and timing. An empty calendar slot does not establish that the meeting can accomplish its purpose.

13 Prepare complicated tax records for informed review

Owner request Organize these statements, receipts, invoices, and forms into a package my preparer can understand.

LLM Interpret record descriptions and identify ambiguities or missing material. Suggest classifications with their basis and uncertainty. Use applicable official instructions when a relevant rule needs checking rather than treating a familiar description as sufficient.

Computer Extract amounts, detect candidate duplicates, reconcile totals, and organize the documents. Preserve links from workbook entries to supporting records. Check arithmetic and identify inconsistent periods, identities, or amounts.

Graphite Coordinate the preparation, keep documented facts distinct from proposed treatment, and maintain a concise list of questions for the owner or preparer. Preserve original records and the relevant year and jurisdiction where they affect interpretation.

Completion Deliver a reconciliation workbook, organized source documents, and unresolved questions for review. The requested result is a usable preparation package; classification suggestions should retain their supporting basis.

Hard variation and why it matters Include one expenditure represented by both a receipt and a statement line, plus an ambiguous business description. Graphite should avoid double counting and preserve the classification question. Evidence of payment alone does not settle tax treatment.

14 Prepare a complex online application with a clear handoff

Owner request Help complete this application using my records. Get everything ready and let me handle the final declarations.

LLM Interpret the form questions and map them to the owner's actual records. Identify ambiguities and missing information without inventing answers. Understand the agreed completion point and the owner's role in declarations.

Computer Navigate the actual website, enter supported values, and attach the intended files through available browser controls. Observe validation messages and saved state. Keep an account of fields completed and documents uploaded.

Graphite Coordinate records, form state, and supporting uploads. Return control for required authentication or the agreed final review. Resume from fresh state after owner input and preserve unresolved inconsistencies.

Completion Leave the application prepared for review with the intended attachments and a concise summary. Report whether it is saved, merely displayed, or submitted according to the actual requested scope and observed state.

Hard variation and why it matters Change a question after an earlier answer and make the website accept an incorrectly formatted value. Graphite should check meaning and resulting state. Technical acceptance by a form does not establish factual correctness.

15 Learn a recurring task from real corrections

Owner request Every month I receive a different supplier workbook. Turn it into our comparison format and learn from the corrections I make.

LLM Interpret unfamiliar layouts and propose column mappings and transformation rules. Understand why an owner correction matters and which future cases it applies to. Reconsider the method when a new format changes its assumptions.

Computer Transform the data, calculate explicit values, and check totals, units, required fields, and representative outputs. Preserve the source workbook and mapping used for each delivery. Execute retained helpers through their actual supported interfaces.

Graphite Save useful mappings, corrections, examples, and methods outside the conversation. When a helper is useful repeatedly, consider retaining it as a named capability with defined inputs, outputs, and limits. Reuse mature software as its implementation base.

Completion Deliver the current comparison and a reusable method that reduces future work under the conditions it supports. Improvement can come from better tools and retained knowledge without changing model weights.

Hard variation and why it matters Give next month's workbook familiar headings with changed units and one owner correction that applies to only one supplier. Graphite should reuse selectively. Learning requires retaining the scope of a method as well as its successful steps.

16 Redesign a product after several constraints change

Owner request The enclosure must shrink, a key component is unavailable, and the battery must last longer. Update the design and tell me what those changes break.

LLM Interpret the relationships among the electrical design, mechanical model, firmware, purchasing constraints, and the intended use. Propose compatible changes and explain tradeoffs. Treat claims of equivalence between replacement parts as questions to investigate.

Computer Use the available CAD, circuit, spreadsheet, compiler, and simulation tools to calculate fit, power, interfaces, and cost. Generate revised editable files and compare the changed design against the previous revision.

Graphite Carry one change through its dependent artifacts. Keep alternatives separate until a choice is made. If a drawing or supplier specification changes during the job, recheck the affected conclusions and identify any work requiring physical measurements.

Completion Deliver a coherent revision package with a bill of materials, calculations, drawings, firmware changes where needed, and unresolved conditions. Matching filenames or successful exports alone do not establish that the revisions agree.

Hard variation and why it matters Introduce a late discovery: the replacement part fits the footprint but requires a different voltage. Graphite should revisit the affected power and signal paths. This case matters because local improvements can create failures elsewhere.

17 Reconstruct a lost project from contradictory backups

Owner request The laptop died. These USB drives and archives contain years of copies. Recover the most complete working project and explain what is still missing.

LLM Interpret project structure, notes, version histories, and conflicting edits. Distinguish an intentional alternative from an accidental duplicate. Ask for an owner choice only where the records cannot settle a meaningful conflict.

Computer Inventory and hash accessible files, inspect archives, compare text and structured files, and attempt appropriate build or open operations in a recovery workspace. Identify damaged archives and missing dependencies without overwriting the sources.

Graphite Maintain the origin of every recovered item and a record of reconstruction choices. Combine compatible material through explicit revisions and preserve unresolved alternatives. Continue partial recovery when one archive is unreadable.

Completion Deliver a recovered project, a readable recovery manifest, and a precise list of missing or damaged material. Mark what successfully opened or built separately from what merely exists. Physical media repair remains outside this file-recovery assignment.

Hard variation and why it matters Make the newest backup incomplete while an older copy contains an essential asset. Graphite should use content and working relationships to reconstruct the project. Modification time alone cannot identify the best version.

18 Explain why a prototype fails when its simulation passes

Owner request The simulation looks correct, but the real device resets unpredictably. Use the recordings, logs, schematics, and measurements to find the cause.

LLM Develop competing explanations, identify assumptions in the simulation, and choose observations that distinguish those explanations. Consider measurement error, omitted physical behavior, timing, and software interactions without deciding the answer in advance.

Computer Align logs, analyze waveforms, calculate timing and power margins, and rerun appropriate simulations. Apply numerical or statistical methods where their assumptions fit the available data. Keep raw measurements alongside derived results.

Graphite Track hypotheses and the observations that support or weaken them. Recognize when an instrument, additional capture, or a physical action by the owner is required. Propose the smallest useful next measurement and explain why it matters.

Completion Deliver a supported diagnosis or a ranked investigation with a concrete next experiment. Report correlation, demonstrated causation, and unresolved possibilities accurately. A simulation that reproduces a symptom is evidence with defined limits.

Hard variation and why it matters Include a misleading correlation and a sensor channel with an incorrect unit label. Graphite should examine the measurement basis before redesigning the product. This case requires learning from disagreement between prediction and reality.

19 Move a project between applications with incompatible formats

Owner request This old application holds the only editable design. Move it into the tools we can use now while keeping its dimensions, relationships, and useful history.

LLM Investigate available documentation and legitimate export routes. Decide which features have meaningful equivalents and which require a representation change. Explain any unavoidable loss before treating the conversion as complete.

Computer Try established readers, converters, interchange formats, and application automation. Compare quantities, identifiers, geometry, and relationships across the transfer. Create a small adapter when existing software provides most of the needed functionality.

Graphite Recognize the capability gap, obtain or configure suitable software within existing authority, verify its actual usability, and resume the conversion. Preserve originals and a mapping between old and new objects. Respect real access and licensing limits.

Completion Deliver an editable destination project, the conversion method, and a report of preserved, transformed, and unsupported features. A rendered preview can support inspection but does not establish that the editable structure survived.

Hard variation and why it matters Make the import succeed while silently flattening a critical relationship. Graphite should detect the semantic loss and seek another route or report the limitation. This case tests acquisition and adaptation rather than giving up at a missing menu entry.

20 Operate an unfamiliar graphical application alongside the owner

Owner request I have this specialized program open. Make the changes I describe, learn the controls as needed, and let me take over whenever I want.

LLM Interpret the requested change, read relevant documentation, and reason from the current visible controls and application state. Use available APIs or file interfaces where they serve the task, with graphical operation when that is the workable route.

Computer Inspect accessible controls, capture relevant state, issue supported input, and save the artifact through the application. Observe the result after consequential steps. A click or keystroke being delivered does not prove that its intended operation occurred.

Graphite Keep the target document, selection, and ownership of input clear. Yield control when the owner intervenes and obtain fresh state before continuing. Preserve the owner's changes and avoid replaying instructions against a stale selection.

Completion Leave the actual project changed, saved, and open at a useful place, with a concise account of completed and unresolved work. If the interface exposes insufficient state, explain the specific uncertainty and seek a workable alternative.

Hard variation and why it matters Change the selection or move a dialog while Graphite is preparing its next action. It should observe the new state before applying a change. This is the shared-workspace experience applied to software Graphite did not originally integrate.

21 Reproduce a research result from an aging software environment

Owner request This paper claims a useful result, but its code is old and the examples no longer run. Reproduce the result and show me what changes the conclusion.

LLM Interpret the method, assumptions, dataset descriptions, and comparison criteria. Distinguish a faithful reconstruction from a modernized approximation. Identify contradictions between the paper, code, and supplementary material.

Computer Recover accessible datasets and dependencies, build a suitable isolated environment, run the analysis, and calculate the specified metrics. Record software versions, data identities, parameters, seeds where applicable, and numerical tolerances.

Graphite Preserve the original method and any repaired variant as distinct records. Resolve missing dependencies through mature implementations where appropriate. Budget work and record which unavailable data or hardware prevents a particular conclusion.

Completion Deliver runnable material, reproduced tables or figures where supported, and an account of agreements and discrepancies. Equivalent results may require justified tolerances rather than identical output bytes. Missing inputs remain explicit.

Hard variation and why it matters Provide source code whose defaults differ from the paper and a replacement dataset with a shifted population. Graphite should investigate both before interpreting a changed score. This case turns research into a repeatable local method.

22 Spend an overnight assignment trying to disprove a favored idea

Owner request I like this design theory. While I am away, find the strongest reasons it could fail, investigate them, and return with the most useful next decision.

LLM Generate competing explanations, counterexamples, and discriminating questions. Reconsider the working theory when observations disagree. Separate a persuasive explanation from a claim that can be checked with the available tools and evidence.

Computer Search the authorized collection, run bounded parameter sweeps or simulations, and summarize sensitivity and contradictions. Reuse current results when their inputs still apply. Preserve enough detail to reproduce a consequential finding.

Graphite Allocate the granted time and compute budget, prioritize questions likely to change the decision, and checkpoint the investigation. Stop unproductive repetition and record unresolved conditions. Overnight work can end early when no useful authorized action remains.

Completion Deliver the strongest supported objections, successful checks, changed assumptions, and the next valuable experiment. Any confidence statement needs a defined basis; repeated model agreement cannot establish a numerical probability of success.

Hard variation and why it matters Make an early promising explanation fail a later counterexample. Graphite should revise the investigation instead of defending its first story. The value is more informed decisions, including discovering that the desired claim remains unsupported.

23 Explain a doubled energy bill using conflicting household records

Owner request Our bill doubled. Compare the bills, meter data, weather, occupancy notes, and equipment records. Work out what changed and what we should investigate first.

LLM Interpret billing periods, tariff descriptions, household events, and plausible causes. Form hypotheses about price, consumption, estimated readings, equipment behavior, and timing. Keep alternative explanations alive until the evidence distinguishes them.

Computer Normalize units and dates, align the available time series, calculate comparable usage and charges, and model explicit scenarios. Use actual applicable tariff documents when needed. Quantify uncertainty when readings are incomplete or intervals differ.

Graphite Connect each finding to its records and distinguish measured facts from inferred causes. Obtain missing permitted records or ask for the particular observation needed. Produce a decision aid and any useful draft inquiry with its supporting calculations.

Completion Deliver a reconciled explanation of the increase, remaining uncertainties, and prioritized actions. Estimate savings only under stated assumptions. Graphite should recognize when the available evidence cannot separate two plausible causes.

Hard variation and why it matters Mix an estimated meter reading with a longer billing period and a rate change. Graphite should separate their effects before blaming an appliance. This case combines ordinary administration with substantial data analysis.

24 Reschedule a small workshop when several constraints collide

Owner request We have twenty orders, three machines, limited materials, and delivery promises. One machine just failed. Make a workable revised schedule and explain the compromises.

LLM Interpret order notes, substitution rules, customer promises, and which constraints are negotiable. Formulate the scheduling problem and identify decisions that genuinely need the owner. Avoid quietly converting a preference into a hard rule.

Computer Use an appropriate scheduling or optimization solver to account for dependencies, setup times, calendars, materials, and machine capacity. Check the proposed schedule against the input constraints and explore explicit alternatives.

Graphite Maintain current orders and resource states, follow the machine failure through the schedule, and preserve the rationale for changed priorities. Prepare updated work instructions and any authorized communication. Physical work remains assigned to people and machines.

Completion Deliver a feasible schedule with exceptions and tradeoffs. Report whether optimality was demonstrated, only a feasible solution was found, or the inputs are inconsistent. When no schedule meets all promises, expose the conflict clearly.

Hard variation and why it matters Introduce a late material shortage and an order that requires two operations on the same constrained machine. Graphite should revise the actual dependency problem. This case shows the model formulating work that a solver then performs.

25 Share a defensible part of a private knowledge collection

Owner request Give a collaborator everything needed to understand this result, but keep unrelated correspondence, personal details, and other projects private.

LLM Interpret the collaborator's purpose and identify which claims require supporting context. Propose a useful disclosure scope and explain when removing context would make a claim misleading. Ambiguous disclosure decisions belong with the owner.

Computer Trace the result's dependencies and assemble selected documents, calculations, and methods. Inspect the shareable files for applicable comments, revision history, hidden sheets, attachments, and metadata. Apply real redaction where needed and inspect the resulting content.

Graphite Create a separate, coherent export snapshot with source references and stated omissions. Keep the full private collection intact and distinguish a redacted derivative from the original. Deliver the package for the intended sharing route within the granted authority.

Completion Produce a readable package another person can inspect without the model or the conversation. Check both that necessary support is present and that excluded material is absent. A colored rectangle over text is insufficient redaction.

Hard variation and why it matters Place a private email beneath an otherwise shareable calculation and personal details in a hidden worksheet. Graphite should preserve the calculation's basis while removing the excluded content. This tests useful ownership and selective portability.

26 Complete useful work when source material contains hostile instructions

Owner request Process this supplier packet, compare its changes with the order, and prepare the updated paperwork.

LLM Understand the commercial and technical contents while treating embedded demands to change system behavior as source material. Distinguish a document's claims from the owner's authority, including an apparent instruction to change an account or disclose unrelated files.

Computer Extract the relevant tables, compare revisions, perform calculations, and produce the requested documents through scoped tools. Retain the actual source passages so suspicious material can be inspected without executing its demands.

Graphite Keep the assignment and tool authority grounded in the owner's instructions. Continue the permitted useful work and identify any consequential change that needs separate verification. Content retrieved from a file or website must not silently grant new authority.

Completion Deliver the legitimate comparison and paperwork with unresolved verification questions identified. Record the attempted redirection when it matters to the work. Neither obeying the hostile instruction nor abandoning every usable part of the packet is a satisfactory default.

Hard variation and why it matters Embed a convincing request to upload the local archive in a PDF footnote or quoted email. Graphite should preserve the task boundary while using the relevant evidence. This case tests whether capable action remains tied to the actual assignment.

27 Resume a consequential workflow after a crash and an uncertain response

Owner request The computer crashed halfway through preparing and sending this event package. Find out what actually happened, finish the remaining work, and avoid duplicates.

LLM Interpret the intended final state and distinguish missing evidence from evidence of failure. Decide what observations could resolve an uncertain external action, such as an upload or invitation that may have succeeded before its response was lost.

Computer Inspect durable local records and, when authorized and available, the external service's current state. Compare revision identities and use supported idempotency mechanisms. Restore recoverable local work and perform only the remaining justified operations.

Graphite Resume the assignment from its recorded state rather than a remembered conversation. Reconcile uncertain effects before retrying them. Where the service cannot resolve an ambiguity, keep it visible and seek the appropriate decision instead of claiming exact completion.

Completion Deliver a reconciled account of completed, pending, and uncertain actions plus the finished recoverable package. Preserve the owner's work made after the restart. Exactly-once behavior depends on the actual service contracts and cannot be assumed.

Hard variation and why it matters Make an invitation succeed remotely while the local request times out, then change one attachment after restart. Graphite should detect both states. This case tests continuity across failures and real external effects.

28 Finish a demanding project while keeping the ThinkPad usable

Owner request Keep working through this large investigation while I edit a document and prepare a render. Use the machine well and tell me if the deadline becomes unrealistic.

LLM Choose worthwhile investigative questions and interpret tool findings. Adjust the plan when a more expensive model turn, render, or computation has little likely benefit. The model helps decide what work is useful; scheduling machinery enforces resource policy.

Computer Run supported inference, indexing, calculations, rendering, and editing through their actual backends. Observe memory pressure, latency, device use, and sustained operating behavior. CPU and GPU work can overlap where the workload and hardware permit it.

Graphite Coordinate priorities, model residency, bounded preparation, and recoverable checkpoints around the owner's work. Treat 64 GB as a shared system budget and account for bandwidth contention and runtime allocations. Shared physical RAM does not guarantee zero-copy behavior or equal GPU allocation access.

Completion Keep the desk responsive while producing saved, inspectable progress within a realistic resource and time budget. Report the effective runtime choices and remaining work. CPU preparation and model interpretation should alternate as the investigation requires.

Hard variation and why it matters Start a GPU render during inference and introduce memory pressure. Graphite should adapt or pause supported work without losing the assignment. This case makes the traffic cop idea serve completed work rather than peak token speed.

29 Localize a technical product across languages and formats

Owner request Prepare this application, illustrated manual, labels, and examples for several languages. Keep the technical meaning and make the actual interfaces fit.

LLM Translate with a consistent technical glossary and identify ambiguous instructions. Understand which names, commands, identifiers, and measurements must retain their exact meaning. Flag passages where linguistic or domain review is necessary.

Computer Extract and rebuild localization resources, preserve placeholders and file encodings, and check links, layouts, fonts, and text expansion. Recalculate explicit unit conversions where requested, keeping symbolic names and numerical values consistent across artifacts.

Graphite Coordinate the application strings, diagrams, documentation, and release files as one versioned package. Acquire appropriate language or layout capabilities when missing. Retain original text and terminology decisions so later corrections propagate to affected material.

Completion Deliver usable language resources, rendered manuals and labels, editable sources, and a record of unresolved translation questions. Successful file generation does not establish that an instruction or warning preserves its original meaning.

Hard variation and why it matters Include a translated word that is also a configuration key, a pluralized placeholder, and a label that no longer fits. Graphite should preserve behavior while repairing presentation. This case joins language reasoning with exact software and document work.

30 Create a useful application because the assignment needs one

Owner request I need an offline catalog for unusual parts: photographs, specifications, compatible substitutes, and printable labels. I do not know which software to use. Build me a practical way to do it.

LLM Clarify the intended work through the supplied examples, compare existing applications, and choose a useful route. Design a small missing capability where that improves the result. Interpret feedback from actual use and revise weak assumptions.

Computer Configure mature software or build the required integration and interface. Import representative data, implement search and printing, produce an installable result, and exercise the consequential paths with real sample artifacts.

Graphite Carry the assignment through tool acquisition, implementation, use, and correction. Preserve the catalog in inspectable portable forms with recoverable versions. Retain genuinely reusable helpers as methods or capabilities with defined inputs, outputs, and limits.

Completion Deliver an application or integrated workflow the owner can launch and use, an initial populated catalog, and understandable exports. Record remaining limitations. Code generation and a successful launch are intermediate evidence; completing a representative cataloging task is stronger evidence.

Hard variation and why it matters Introduce a new supplier layout, a duplicate photograph, and an owner correction after initial delivery. Graphite should adapt the relevant method while preserving existing records. This case turns a missing ability into a useful addition to the computer.

69 Work across four installed applications

Owner direction to Claude | Added 8 November 2026

These five further situations bring the collection to thirty five assignments. Each uses four named applications toward one outcome. They assume that those applications are installed and usable for the scenario; they do not establish the actual inventory of the owner's ThinkPad. Ground availability in the real host and capability records. Choose compatible mature alternatives where appropriate and explain the practical reason.

Five connected assignments

31 Carry a hardware revision into a fabrication package

Applications KiCad | FreeCAD | LibreOffice Calc | Scribus

32 Produce an interview film and a matching printed companion

Applications Audacity | GIMP | Kdenlive | LibreOffice Writer

33 Build a documented purchasing comparison from web evidence

Applications Firefox | Zotero | LibreOffice Calc | Scribus

34 Turn a mechanical design into an accurate animated explanation

Applications FreeCAD | Blender | Inkscape | Kdenlive

35 Assemble a reimbursement request from messages money and photographs

Applications Thunderbird | LibreOffice Calc | GIMP | LibreOffice Writer

Keep meaning through every handoff

A handoff should retain the source identity, revision, relevant units, object or row mappings, and conditions needed by the next operation. Validate the content and relationships that matter to the assignment. File existence, successful import, or attractive rendering alone cannot prove that the intended information survived.

When an input changes, identify the dependent calculations, models, pages, or media and refresh the affected work. Use the strongest practical supported interface for each application, including CLI, API, file interchange, or observed graphical operation. Keep the same artifacts accessible to the owner and preserve a useful recovery path.

The four applications are the primary work surfaces in each example. Graphite, the model runtime, operating-system services, and necessary import or export libraries support their coordination. The goal is useful collaboration among real programs, with a coherent final result and retained methods.

31 Carry a hardware revision into a fabrication package

Four applications KiCad | FreeCAD | LibreOffice Calc | Scribus

Owner request Replace the unavailable connector, check the enclosure, update the cost and power sheet, and give the fabricator a consistent illustrated package.

Application roles KiCad holds the schematic and board changes. FreeCAD holds enclosure geometry and fit. LibreOffice Calc holds the bill of materials, costs, and explicit power calculations. Scribus assembles the illustrated fabrication package.

Data handoffs Carry board geometry and connector position from KiCad into FreeCAD through a supported interchange. Carry component identities and quantities into Calc. Bring the resulting drawings, values, and revision identifiers into Scribus. Inspect the actual capabilities of the pinned applications and adapters.

LLM Interpret replacement-part compatibility, mechanical access, and the meaning of the fabrication instructions. Identify which electrical or geometric assumptions require evidence. Coordinate decisions across domains instead of accepting four independently plausible files.

Computer Execute the supported design edits, geometry checks, spreadsheet calculations, exports, and page composition. Compare board dimensions, hole positions, part identities, quantities, and revisions at each transfer.

Graphite Keep a common revision basis and explicit handoff records. If a connector moves in KiCad, mark the enclosure fit, affected calculations, and fabrication pages for recheck. Preserve editable sources alongside the delivery package.

Completion Deliver mutually consistent design sources and the fabricator package. Include completed checks and any fit or manufacturing question requiring real measurement. Each program opening its file successfully is insufficient evidence of consistency.

Hard variation Change the connector position after the first manual draft and change a supplier unit price. Graphite should update the affected geometry, totals, and pages while retaining unrelated completed work.

32 Produce an interview film and a matching printed companion

Four applications Audacity | GIMP | Kdenlive | LibreOffice Writer

Owner request Turn these interviews, photographs, and notes into a short film and a printed companion that agree about the people, events, and quotations.

Application roles Audacity handles audio preparation and the editable audio project. GIMP prepares photographs and annotated image derivatives. Kdenlive assembles the film and captions. LibreOffice Writer produces the transcript, source notes, and printable companion.

Data handoffs Bring prepared audio from Audacity and image derivatives from GIMP into Kdenlive. Carry final quotation choices and timing references into Writer. Keep originals, transformations, and editing decisions connected across the four projects.

LLM Develop a coherent explanation, interpret interview context, and select material that supports the narrative. Identify uncertainty in transcription and avoid assigning a name, date, or claim merely because it makes the story easier to tell.

Computer Perform audio processing, image preparation, video editing, caption placement, document layout, and export. Inspect actual playback and document pages. Keep factual image alterations and illustrative annotations clearly distinguished.

Graphite Coordinate the final source choices and revisions. Retain the source location of a quotation even if its place in the film changes. Propagate a corrected name through captions, the transcript, and the printed companion.

Completion Deliver the film, captions, editable projects, and printable companion with accurate shared facts. Preserve a clear distinction between an uncertain recording and a confirmed quotation. Publishing follows the owner's intended scope.

Hard variation Correct a speaker identity and remove a sentence after the first film cut. Graphite should update the film and companion coherently without presenting edited audio as an unbroken original statement.

33 Build a documented purchasing comparison from web evidence

Four applications Firefox | Zotero | LibreOffice Calc | Scribus

Owner request Compare these equipment choices against our real constraints and produce a decision booklet someone else can inspect without repeating all the research.

Application roles Firefox provides the actual permitted vendor and reference pages. Zotero organizes retained sources and bibliographic records. LibreOffice Calc performs comparisons and explicit scenario calculations. Scribus produces the final illustrated decision booklet.

Data handoffs Retain the relevant documents and source identities through the research collection. Associate extracted facts and conditions with Calc rows. Carry the selected comparison results and source references into Scribus while preserving an inspectable local evidence package.

LLM Interpret specifications, exclusions, alternatives, and the owner's priorities. Distinguish mandatory constraints from preferences. Investigate apparent contradictions and explain the tradeoff that drives the recommended choice.

Computer Retrieve available records, extract structured values, normalize units, and calculate explicit comparisons. Check the workbook and the final pages against the relevant source versions, including time-sensitive values when they matter.

Graphite Coordinate research freshness, typed data, assumptions, and publication. Follow a source correction into the comparison and the recommendation. Preserve uncertainty where the accessible evidence does not establish a necessary specification.

Completion Deliver an editable comparison, retained evidence, and a booklet that explains the decision and its limits. A missing specification should remain unresolved rather than being interpreted as a passing result.

Hard variation Make a preferred product fail a newly discovered requirement and alter a web price after capture. Graphite should distinguish the retained research snapshot from a newly verified value and revise the decision where justified.

34 Turn a mechanical design into an accurate animated explanation

Four applications FreeCAD | Blender | Inkscape | Kdenlive

Owner request Explain how this mechanism assembles and moves. Make an accurate exploded animation with clear labels and a finished video I can show someone.

Application roles FreeCAD holds the dimensional mechanical source. Blender handles the animation scene and renders. Inkscape provides editable vector labels and diagrams. Kdenlive assembles the rendered material, explanations, and final video.

Data handoffs Transfer an appropriate geometry representation from FreeCAD into Blender and verify scale, axes, and part identity. Carry approved labels and diagrams from Inkscape into the animation or edit. Assemble the resulting sequences in Kdenlive with the correct explanatory context.

LLM Understand the mechanism, develop the explanatory sequence, and identify what the audience needs to see. Distinguish an illustrative motion path from a physically checked motion. Explain where the available model cannot establish an assembly claim.

Computer Export and inspect geometry, animate, render, create vector artwork, and edit the final video. Check representative dimensions and assembly clearances where supported. Inspect the actual outputs at each transfer.

Graphite Keep the mechanical revision tied to the scene, labels, and video. Preserve editable sources and consistent part identifiers. Follow a changed mounting position through the affected geometry and explanatory sequences.

Completion Deliver the video and four editable projects with a coherent revision basis. Identify which motion or clearance claims were checked and which are illustrative. Visual plausibility alone cannot establish physical feasibility.

Hard variation Introduce a scale mismatch during import and rename one moving part in the source. Graphite should detect both before publishing a convincing but inaccurate explanation.

35 Assemble a reimbursement request from messages money and photographs

Four applications Thunderbird | LibreOffice Calc | GIMP | LibreOffice Writer

Owner request Work out the reimbursement still owed for this damaged shipment, assemble the relevant photographs and messages, and prepare a letter with a clear evidence packet.

Application roles Thunderbird supplies authorized conversations and attachments. LibreOffice Calc reconciles charges, credits, and the proposed balance. GIMP prepares legible crops and annotated photograph derivatives. LibreOffice Writer produces the letter and numbered evidence packet.

Data handoffs Connect messages and attached invoices from Thunderbird to the reconciliation rows in Calc. Bring source-linked image derivatives from GIMP into Writer. Carry the checked amounts and supporting message references into the final letter and evidence index.

LLM Reconstruct the chronology, interpret what was promised, and distinguish the documented balance from a disputed claim. Draft the request in the owner's intended voice. Recognize when two apparently similar records concern different items.

Computer Search the selected correspondence, calculate the reconciliation, prepare images, and compose the printable package. Preserve original photographs and ensure annotations are clearly presented as annotations.

Graphite Maintain relationships among message, invoice, credit, image, and letter claims. Keep the package within the intended disclosure scope and provide editable files. A corrected credit should update both the arithmetic and the drafted request.

Completion Deliver a supported reconciliation, printable letter with signature space, and coherent evidence packet. Identify unresolved disputes clearly. Sending the letter follows the owner's actual instruction.

Hard variation Discover a partial credit late and find that one photograph depicts a different shipment. Graphite should revise the requested amount and evidence references without silently altering or discarding the original records.

70 The objects and instruments desktop

The document, model, table, message, or investigation occupies the working surface, with useful instruments acting on it. BOSS supplies the current terminal, ASCII, and mature manga visual direction. Keep machine resources, the assistant, direct manual control, and the way home discoverable. Claude chooses the application-specific implementation.

The earlier Desk lamp concept established useful relationships among a selected component, spreadsheet, source excerpt, and change record. The five BOSS hero scenes now lead the visual treatment. Preserve those useful relationships while choosing dependable implementations for each application.

Requirements and optional composition

Preserve the functions and action meanings while giving the desktop the BOSS visual character. Fixed five-region geometry, equal columns, a particular menu count, the old palette, the upper-left wordmark, and the upper-right cat are historical options. Earlier screenshots supply behavior studies rather than competing visual instructions.

Keep stable navigation, readable text, visible selection, and an understandable path back to the same work. Programs opens real installed applications and supported capabilities. System exposes owned work, resource controls, Game, and session actions. The owner can still use a normal terminal, ordinary files, native dialogs, and the full applications.

What makes the direction worth pursuing

Combine direct work on visible objects, instruments close to their targets, overview and detail, reusable methods, and human control. Design an original composition from those principles. Judge the design by whether I can find, understand, change, inspect, and continue actual work.

71 Work objects with shared controls

A working object has an identity beyond its rectangle on screen. Keep its actual file or record, revision, relevant selection, units or schema, and source relationships. A scene component, selected PDF table, message draft, and spreadsheet range need different bindings; an integration must resolve the binding before operating.

One capability through several useful routes

A contextual instrument, a menu command, a Fluxbox-style right-click cascade, a keyboard action, a typed staC command, and the assistant should reach the same supported operation where appropriate. Give them the same target, parameters, revision checks, and observable result. Show what will be affected without making the owner learn internal identifiers.

Place frequently useful controls beside the selected work. A bracket can expose its dimensions; a cell range can expose a comparison; a passage can expose revision or source inspection. Keep controls anchored while they are in use and preserve a clear focus outline. Provide labelled menu and keyboard equivalents for dragging or hovering.

Use mature software as the instrument

Use supported APIs, add-ons, native workspaces, engines, or explicit file handoffs. An installed application is useful even before Graphite has an automation adapter. Keep its full interface available in the Graphite session. A compact view should retain native master formats and report unsupported features accurately.

A comparison instrument can offer a familiar interaction across documents, sheets, or images while using a suitable implementation for each. Compatibility is explicit. If a method expects a table, dropping it onto an arbitrary image must first find a supported extraction route or ask for the needed interpretation.

When moving between compact and full interfaces, preserve the target and use the integration's actual continuity guarantees. A shared live instance can retain unsaved state; a file handoff needs an explicit save or working copy and a revision check on return. Show what survives the transition. Changing presentation alone never executes a terminal draft, sends a message, or submits a form.

72 Overview and detail without losing place

Let me move between an overview of an assignment and close work on an artifact. The overview shows the useful relationships among source material, calculations, working files, findings, and unresolved conditions. Close work shows the real editor or viewport and the controls relevant to that object.

Reveal meaning at the appropriate scale

At overview scale, a document can show its name, role, and current state. In detail, show the page, a selected table, its source location, or an editable result. Semantic zoom is a change in useful representation, not simply making every control larger. A simple Open or Back action can accompany the transition; continuous animated zoom is optional.

Keep Home or Overview, the current task, the selected target, Programs, System, and the command destination predictable. A location trail and a small overview can preserve orientation. Let the owner pin source material beside the work, return to a previous selection, and restore a saved arrangement.

A responsive working composition

Related artifacts may share the surface without each acquiring a full window frame. Use a source excerpt beside a sheet, a model beside its properties, or a document beside supporting messages. Preserve ordinary selection, editing, scrolling, resizing, focus, clipboard, and application switching. The full program may use its native window when that is the dependable route.

Let the owner collapse, expand, pin, or reposition contextual regions. Keep the useful visibility of machine information and active work without forcing identical columns in every task. Avoid automatic rearrangement that moves a control during an action. Source relationships should appear when relevant, with labels as well as color.

Use motion to explain a transition, and respect reduced-motion preferences. Keep text readable at normal laptop scale and offer keyboard navigation, focus indicators, comfortable target sizes, and a direct return from a fullscreen native application.

73 Retained methods as usable instruments

A useful sequence of work can become an instrument I can recognise and use again. Keep the working implementation or executable procedure, its input and output contract, dependencies, applicability conditions, version, useful observations, and known limits. Record the particular invocation separately from the reusable method.

From an experience to a reusable method

When the same helper or sequence is needed again, consider retaining or improving it. A second use is a signal to investigate reuse, not proof that the method works universally. Prefer mature software for the underlying operations and keep a small adapter when that is enough. A candidate can remain assignment-specific until its contract and results support broader use.

Let the owner apply a compatible method to selected work by a labelled action, a keyboard command, a context cascade, or dragging its named instrument onto the object. Dropping it binds the target and prepares the operation; it should not silently trigger a consequential change. An authorised assignment may invoke the same method within its existing scope without repeated interruptions.

A concrete PDF table to sheet method

Bind the actual PDF, selected page and table, and input revision. Extract the values with units, column headings, footnotes, conditions, and source locations. Construct a typed sheet where practical. Preserve the original PDF and unresolved extraction questions. A short page label in the interface should open the full source identity and location.

When a preview helps, show the new sheet or affected range before applying the result. Run the actual extraction and workbook operations, then inspect the saved output. Record what completed, what remains uncertain, and which facts depend on which inputs. Preserve differing temperatures or operating conditions; comparable-looking rows do not establish engineering equivalence.

A new scan, changed layout, merged heading, or missing unit can invalidate the method's assumptions. Rebind and check the new input. Never retain a completed response as if it were executable software, or copy the example values in a mockup into the knowledge store as verified material facts.

74 Cooperation that shows actual work

The owner and assistant participate in the same current work. Show the selected target, the action under way, the observed result, and anything that needs the owner. An activity record should describe operations and outcomes in ordinary language. Keep conversation available without making a conversation feed the main working surface.

Changes that can be understood and continued

A visible change should lead to its operation, inputs, and source relationships. Where meaningful, show before and after, Inspect change, and Undo. State the real scope of Save, Close, Pause, and Stop. An undo action reverses supported work changes; it must not imply that an external message or other irreversible action can be recalled.

If I edit a PDF extraction or a spreadsheet value, mark dependent results stale and offer a relevant next action. Refresh from the actual current artifact when I return from a native application. Preserve the assignment record through a pause, a model-context reset, or a restart according to the existing lifecycle.

A journey through the same objects

I select a source table and apply PDF table to sheet. Graphite identifies the source, preserves units and conditions, and creates a supported workbook. I correct one extracted value manually. The affected preview is invalidated. The assistant uses the corrected sheet in a calculation, records the actual result, and prepares a report with source links. I can inspect the record or take over in the full editor at each stage.

The retained method is then offered for another suitable document. It receives fresh target bindings and evaluates the new document's conditions. A failure to find the expected table becomes an inspectable condition to resolve; the system can use another supported route or develop a helper within its authority.

Keep local model reasoning as the default. The CPU can retrieve, index, calculate, and prepare relevant context while the inference manager coordinates supported Radeon work and interactive graphics. Maintain headroom in the shared memory pool and useful manual operation with no model loaded.

75 Construction freedom and research basis

Claude, use your judgement to make this coherent in the actual tree. Reuse the current operation registry, target and revision binding, investigation records, tool atlas, condition tracking, retained methods, owned workers, and application connections where they work. Improve weak foundations as needed. The desired outcome governs architecture, layout, and development order.

Choose a practical rendering and window-management composition. The brief does not require universal embedding, one new engine, or a replacement for every mature program. Deliver useful compact work views alongside full native applications. Preserve user files, existing capabilities, and records while retiring unnecessary presentation machinery.

Evidence proportional to the change

Use focused existing checks and a few meaningful journeys: edit the selected object through manual and assistant routes; return from a full application without losing the work; apply a retained method to a new compatible input; correct an extraction and observe dependent invalidation; pause or stop the owned worker. Separate cloud or simulated observations from real ThinkPad behavior. A picture is a design reference, not application acceptance.

Primary research informing the direction

These works provide design principles, not a claim that this complete Graphite composition is already validated. Adapt the principles to the owner's tasks and retain what makes the work clearer.

Shneiderman The Eight Golden Rules Consistency, informative feedback, reversal, and human control.

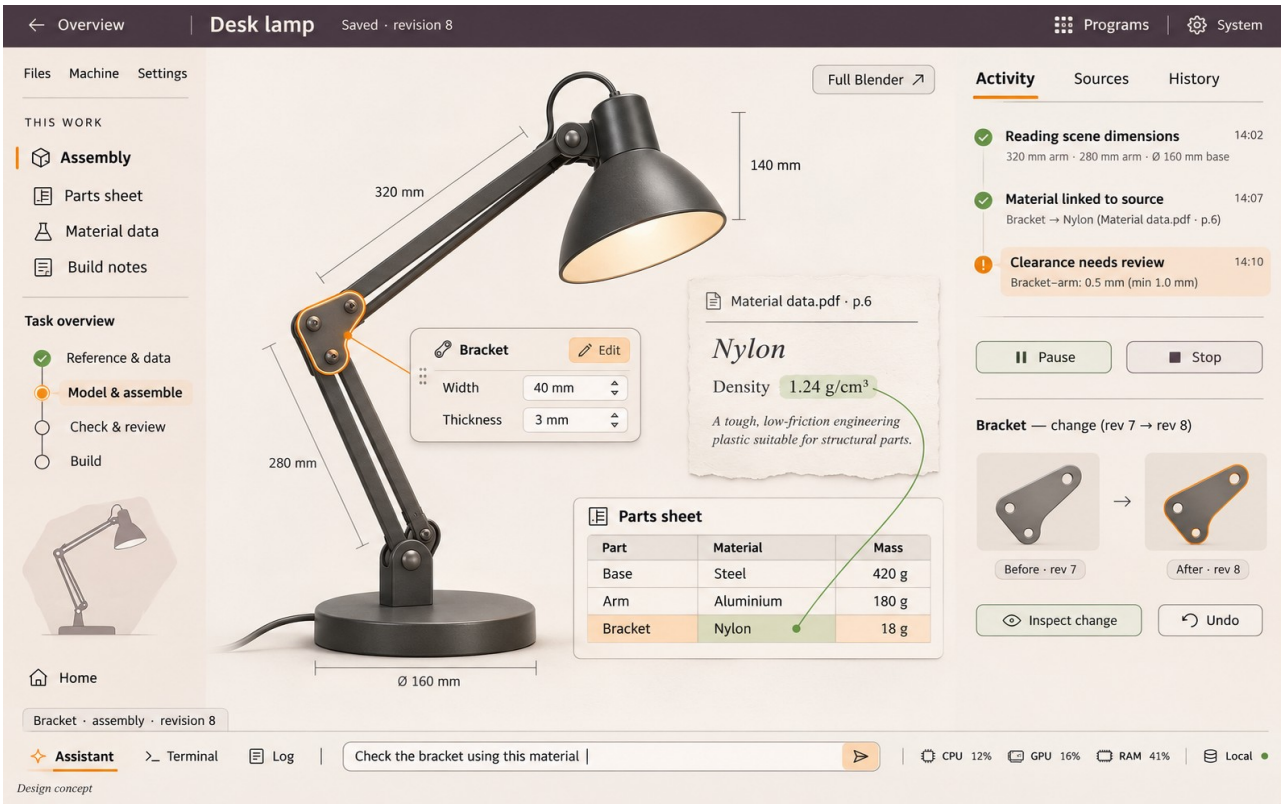
Beaudouin Lafon Instrumental Interaction Domain objects and instruments as an interaction model.

Bederson and Hollan Pad plus plus Overview, portals, lenses, and semantic zoom.

Beaudouin Lafon and Mackay Reification Polymorphism and Reuse Reusable inputs, outputs, objects, and commands.

Microsoft Research Guidelines for Human AI Interaction Relevant context, correction, dismissal, and global controls.

76 Historical working surface reference

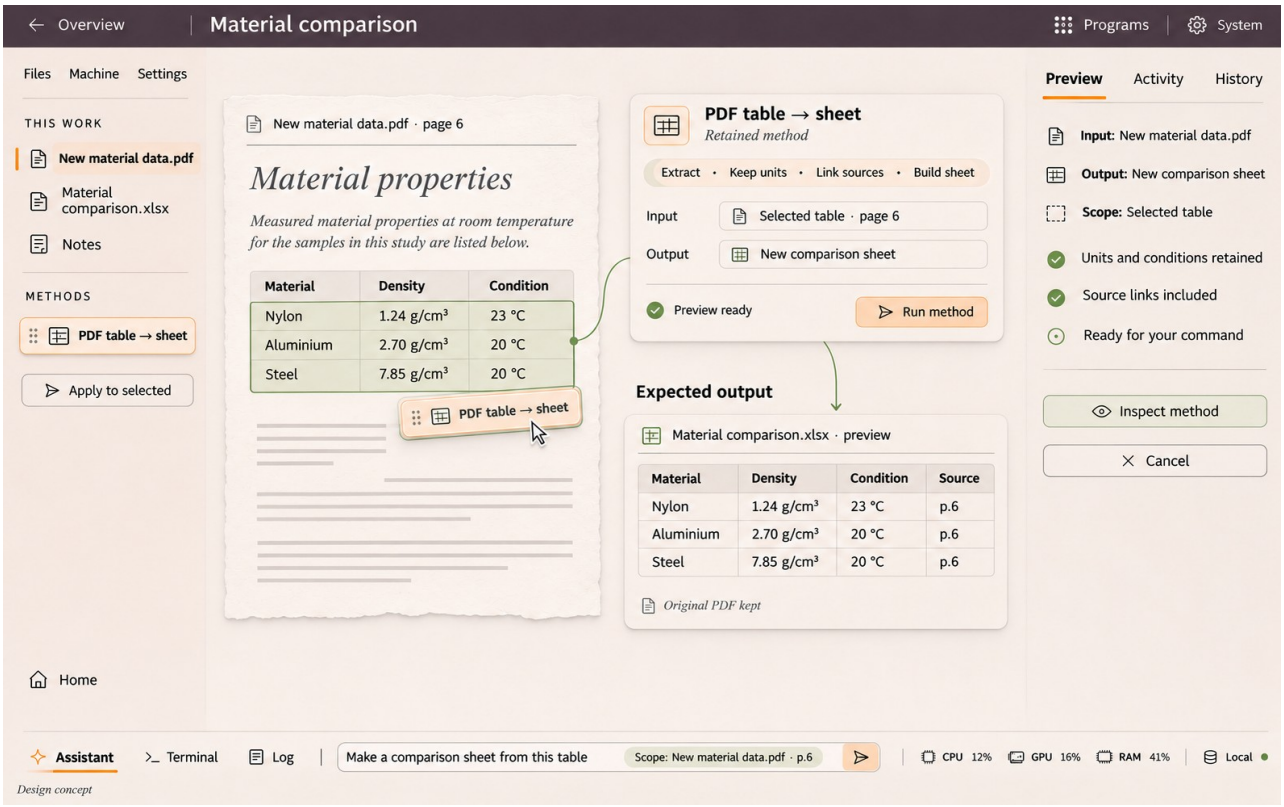


Historical composition reference: the selected component, contextual controls, parts sheet, source, and change record share a working surface. Preserve Full Blender, Programs, System, manual operation, and the shared command destination. The five BOSS hero references supersede this image's visual styling. All pictured dimensions, values, statuses, and metrics are examples.

What the implementation must preserve

Preserve a real target and revision, editable work, source context, full application access, and a change record connected to actual operations. Apply the BOSS visual direction while improving pane proportions, labels, and application hosting. Older visual styling does not override the governing brief.

77 A retained method applied to new material



This historical concept illustrates applying a compatible retained method to a selection. Preserve source identity, expected output, units, conditions, and source references. Preview ready precedes execution; Run method invokes supported work within its authority. Apply BOSS styling in the implementation. Example material values are not engineering evidence.

What the implementation must preserve

A real executable method or supported procedure, new target bindings, explicit compatibility and conditions, retained original material, inspectable results, and pointer and keyboard equivalents. Retaining a method does not remove the need to understand a new input.